Radial Basis Function-generated Finite Differences: A Mesh-free Method for Computational Geosciences

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Abstract

Radial basis function generated finite differences (RBF-FD) is a mesh-free method for numerically solving partial differential equations (PDEs) that emerged in the last decade and has shown rapid growth in the last few years. From a practical standpoint, RBF-FD sprouted out of global RBF methods, which have shown exceptional numerical qualities in terms of accuracy and time stability for numerically solving PDEs, but are not practical when scaled to very large problem sizes because of their computational cost and memory requirements. RBF-FD bypass these issues by using local approximations for derivatives instead of global ones. Matrices in the RBF-FD methodology go from being completely full to 99% empty. Of course, the sacrifice is the exchange of spectral accuracy from the global RBF methods for high-order algebraic convergence of RBF-FD, assuming smooth data. However, since natural processes are almost never infinitely differentiable, little is lost and much gained in terms of memory and runtime. This article provides a survey of a group of topics relevant to using RBF-FD for a variety of problems that arise in the geosciences. Particular emphasis is given to problems in spherical geometries, both on surfaces and within a volume. Applications discussed include non-linear shallow water equations on a sphere, reaction-diffusion diffusion equations, global electric circuit, and mantle convection in a spherical shell. The results from the last three of these applications are new and have not been presented before for RBF-FD.

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1 Introduction to the concept of radial basis functions (RBF)

The motivation for the RBF method originated with R.L. Hardy asking the question, 'Given a set of sparse *scattered* data, $\{f_j\}_{j=1}^N$, at the node locations $\{\mathbf{x}_j\}_{j=1}^N \subset \mathbb{R}^d$, can an interpolant be constructed that adequately represents the unknown surface?' [32]. It was first shown by Mairhuber [39] that, in more than one dimension, interpolation is not well-posed when using an expansion of basis functions, $\{\psi_j(\mathbf{x})\}_{j=1}^N$, $\mathbf{x} \in \mathbb{R}^{d\geq 2}$, that are independent of the node locations. That is, there exists an infinite number of node configurations that will yield a singular interpolation problem. Hardy bypassed this singularity problem by constructing the interpolant from a basis function set consisting of translates of a single radially symmetric function with one centered at each data location. By giving up orthogonality, well-posedness of the interpolant and its derivatives for *any* set of distinct scattered nodes in *any* dimension is guaranteed. Although unconditional non-singularity of the interpolation problem was known early in some special cases [5; 47], the proof in 1986 of guaranteed non-singularity for multiquadric (MQ) RBFs [40] accelerated the further development and acceptance of RBFs. Pioneering work by M.J.D. Powell and his collaborators at University of Cambridge played also a major role in the early history of RBFs [43].

Piecewise smooth RBFs feature a jump in some derivative and thus can only lead to algebraic convergence. For example, the radial cubic $|r|^3$, where $r = \|\mathbf{x} - \mathbf{x}_j\|$ is the Euclidean norm, has a jump in the third derivative at $\mathbf{x} = \mathbf{x}_j$, leading to fourth order convergence in 1-D, with the order of convergence increasing as the dimension increases (c.f. [43]). On the other hand, interpolating with infinitely smooth RBFs, such as $\sqrt{1+(\varepsilon r)^2}$, $\exp(-(\varepsilon r)^2)$, and $1/(1+(\varepsilon r)^2)$ will lead to spectral convergence [38; 62]. Table 1 shows commonly used RBFs, noting that infinitely smooth RBFs depend on a shape parameter ε. It was first shown by Driscoll and Fornberg [10] that, in 1-D, in the limit of $\varepsilon \to 0$ (i.e. flat RBFs) the RBF methodology reproduces pseudospectral methods (PS) if the nodes are accordingly placed (i.e. equispaced nodes for Fourier methods, Gauss-Chebyshev nodes for Chebyshev methods, etc.). Similarly, on the surface of a sphere, Fornberg and Piret [16] showed that, in the limit of $\varepsilon \to 0$, RBFs reproduce spherical harmonics in the sense that they span an equivalent space for any scattered node set. Later in the chapter, algorithms of how to compute in the near flat RBF limit will be discussed.

Table 1: Some common choices for radial functions

Type of basis function	**Radial function $\phi(\mathbf{r})$**
Piecewise smooth RBFs	
Generalized Duchon spline (GDS)	$r^{2m}\log r,\ m \in \mathbb{N}$
	$r^{2m},\ m > 0$ and $m \notin \mathbb{N}$
Matern	$\frac{2^{1-m}}{\Gamma(m)} r^m K_m(r), m > 0$ (Bessel K-function)
	Special cases: e^{-r}, $e^{-r}(1+r)$ for $m = \frac{1}{2}, \frac{3}{2}$.
Compact support ('Wendland')	$(1-r)_+^m p(r)$, p certain polynomials, $m \in N$
Infinitely smooth RBFs	
Gaussian (GA)	$e^{-(\varepsilon r)^2}$
Multiquadric (MQ)	$\sqrt{1+(\varepsilon r)^2}$
Inverse Multiquadric (IMQ)	$1/\sqrt{1+(\varepsilon r)^2}$
Inverse Quadratic (IQ)	$1/(1+(\varepsilon r)^2)$
Sech (SH)	sech εr
Bessel (BE) ($d = 1, 2, \ldots$)	$J_{d/2-1}(\varepsilon r)/(\varepsilon r)^{d/2-1}$

The primary focus of this article is on the application of RBF generated finite differences (RBF-

FD) for numerically solving PDEs, with specific emphasis on PDEs in spherical geometries that arise in the computational geosciences. Overviews of the theoretical aspects of RBF interpolation and approximation can be found in [7; 56], while an overview of more numerically focused aspects can be found in [12]. The interested reader can find more general approximation theoretic results in spherical geometries with applications to the geosciences in [2; 24; 25; 26]. The latter three of these monographs even include vectorial and tensorial generalizations of RBFs and their applications.

2 RBF-generated finite difference (RBF-FD) approximations

Before delving into differentiation, interpolation is discussed as it can be considered a zeroth order differentiation. RBF interpolation is based on a linear combination of translates of a single radially symmetric function, $\phi(\|\mathbf{x}-\mathbf{x}_k\|)$, that collocates the data $\{f_k\}_{k=1}^N$ at the nodes $\{\mathbf{x}_k\}_{k=1}^N$, $\mathbf{x}_k \in \mathbb{R}^d$, and is given by

$$s(\mathbf{x}) = \sum_{k=1}^{N} c_k \phi(\|\mathbf{x}-\mathbf{x}_k\|), \tag{1}$$

where $\|\cdot\|$ denotes the ℓ_2 norm. The expansion coefficients, c_k, can be found by inverting the matrix, A, in (2)

$$\underbrace{\begin{bmatrix} \phi(||\mathbf{x}_1-\mathbf{x}_1||) & \phi(||\mathbf{x}_1-\mathbf{x}_2||) & \cdots & \phi(||\mathbf{x}_1-\mathbf{x}_N||) \\ \phi(||\mathbf{x}_2-\mathbf{x}_1||) & \phi(||\mathbf{x}_2-\mathbf{x}_2||) & \cdots & \phi(||\mathbf{x}_2-\mathbf{x}_N||) \\ \vdots & \vdots & \ddots & \vdots \\ \phi(||\mathbf{x}_N-\mathbf{x}_1||) & \phi(||\mathbf{x}_N-\mathbf{x}_2||) & \cdots & \phi(||\mathbf{x}_N-\mathbf{x}_N||) \end{bmatrix}}_{\text{Interpolation matrix A}} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_N \end{bmatrix}. \tag{2}$$

Since RBFs only depend on a scalar distance (defined by the ℓ_2 norm), the form of (2) is independent of *coordinate system*, *dimension* and *domain geometry*. As an example, even if node locations on a sphere are given in spherical coordinates, no such grids need to be used. Distances are measured straight through the sphere and not along great arcs. That is, the argument of the RBF centered at, (x_1, y_1, z_1) and evaluated at (x_2, y_2, z_2) is:

$$\begin{aligned} r =& \sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2} = \sqrt{2(1-\mathbf{x}_2^T\mathbf{x}_1)} \\ =& \sqrt{2(1-\cos\theta_2\cos\theta_1\cos(\lambda_2-\lambda_1)-\sin\theta_2\sin\theta_1)}. \end{aligned}$$

where $\theta_{1,2}$, $\lambda_{1,2}$ are the respective latitude and longitude of the nodes.

The RBF differentiation matrix (DM), D_N, is derived by applying the analytic derivative operator L of the RBF interpolant and evaluating it at the desired node locations. As an example on a sphere, to approximate L at $(\lambda,\theta) = (\lambda_i, \theta_i)$

$$\begin{aligned} Ls(\lambda_i,\theta_i) =& \sum_{k=1}^{N} c_k \underbrace{\left[L\phi_k(r)\right]\big|_{(\lambda,\theta)=(\lambda_i,\theta_i)}}_{\text{Components of } B} \qquad (i=1,\ldots,N) \\ =& B\boldsymbol{c} \\ =& (BA^{-1})\mathbf{f} \\ =& D_N\mathbf{f}, \end{aligned} \tag{3}$$

where $\mathbf{f}$ contains the N discrete data values at the node locations, $\boldsymbol{c}$ contains the N discrete expansion coefficients and is formally given by $\boldsymbol{c} = A^{-1}\mathbf{f}$, where A^{-1} is the inverse of the RBF interpolation matrix defined in (2).

The above derivation in (3) is for global RBF DM. That is, to calculate the discretized derivative operator at one node, all the nodes in the domain are used. This is analogous to a spectral collocation method. Thus, for smooth data and using infinitely smooth RBF, spectral convergence can be expected. Deriving RBF-FD DMs is very similar, except that for each stencil, only n nodes out of the N nodes in the domain are used. Similar to regular finite differences, as the stencil size n increases so does the rate of algebraic convergence. Now, the differential operator L is approximated at the node $\mathbf{x}_c$ by a linear combination of the function values, u_k, at the neighboring $n-1$ node locations, $\mathbf{x}_k$. In other words $\sum_{k=1}^{n} a_k u_k = Lu_c$. The differentiation weights, a_k, are calculated by enforcing that this linear combination should be exact for RBFs, $\{\phi(\|\mathbf{x}-\mathbf{x}_k\|)\}_{k=1}^{n}$, centered at each of the node locations $\{\mathbf{x}_k\}_{k=1}^{n}$ (classical FD would enforce that it be exact for polynomials instead). It has also been shown through experience and studies [15; 19; 60] that better accuracy is gained by the interpolant being able to reproduce a constant. Hence, the constraint $\sum_{k=1}^{n} a_k = L1|_{\mathbf{x}=\mathbf{x}_c} = 0$ is added. Combining these constraints, the solution for the RBF-FD weights can be determined from the following linear system:

$$\begin{bmatrix} \phi(\|\mathbf{x}_1-\mathbf{x}_1\|) & \phi(\|\mathbf{x}_1-\mathbf{x}_2\|) & \cdots & \phi(\|\mathbf{x}_1-\mathbf{x}_n\|) & 1 \\ \vdots & & \ddots & \vdots & \\ \phi(\|\mathbf{x}_n-\mathbf{x}_1\|) & \phi(\|\mathbf{x}_n-\mathbf{x}_2\|) & \cdots & \phi(\|\mathbf{x}_n-\mathbf{x}_n\|) & 1 \\ 1 & 1 & \cdots & 1 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \\ a_{n+1} \end{bmatrix} = \begin{bmatrix} L\phi(\|\mathbf{x}-\mathbf{x}_1\|)|_{\mathbf{x}=\mathbf{x}_c} \\ \vdots \\ L\phi(\|\mathbf{x}-\mathbf{x}_n\|)|_{\mathbf{x}=\mathbf{x}_c} \\ 0 \end{bmatrix}, \quad (4)$$

where a_{n+1} is not actually used in RBF-FD approximation after the system is solved. Solving this system once gives one row of the RBF-FD DM . For a total of N nodes on the sphere, there will be N linear systems of size $(n+1)\times(n+1)$ to solve, resulting in a preprocessing cost of $\mathrm{O}(n^3 N)$. Each subsequent matrix-vector multiplication will then cost only $O(nN)$ operations. In high resolution computations, $N >> n$, and thus the cost for each time step of the RBF-FD method is $O(nN)$. This results in a significant speed-up from global RBFs that require $\mathrm{O}(N^3)$ operations to create the DMs and $\mathrm{O}(N^2)$ to time-step. In general, a kd-tree algorithm is used to find the $n-1$ nearest neighbors when calculating the differentiation weights for the stencils. MATLAB conveniently has a kd-tree algorithm called 'knnsearch' in its statistical toolbox. RBF-FD matrices are very sparse, generally about 99% empty. Thus, in order to reduce the bandwidth the matrix as well as indexing the entries effectively in memory, a Cuthill-McKee algorithm is applied to the RBF-FD DM via the MATLAB statement 'symrcm'.

2.1 Limit of flat basis functions

With ε available as a free parameter, an obvious numerical test is to see how the choice of it influences the accuracy that is obtained. A typical experiment is shown in Figure 1. The error is observed to be rapidly decreasing with ε when the calculation suddenly breaks down due to the ill-conditioning of *RBF-Direct* - evaluation of (2) followed by (1). This may suggest that a tradeoff will be required between accuracy and numerical conditioning (described as an '*uncertainty principle*' in [45]) . It was however realized shortly afterwards that the RBF interpolation problem itself actually does not become ill-conditioned in the flat basis function limit [10; 20]. Instead, it is only the RBF-Direct algorithm that then becomes an ill-conditioned numerical procedure for a problem that remains well conditioned. Several well conditioned numerical algorithms were subsequently developed. Using these, it was soon found that RBF errors often become extremely low before

leveling off or bouncing back up, cf. Figure 2. Sometimes, the most accurate ε-range can be reached already with RBF-Direct, while at other times it requires a stable algorithm (or more costly extended precision arithmetic).

If the nodes are lattice-like, it can happen that the RBF interpolant diverges as $\varepsilon \to 0$ (although never in the GA case [46]). For node sets with some irregularity, the interpolant will in the flat limit take the form of a multivariate polynomial [10; 20]. One reason that ε small often is better than $\varepsilon \to 0$ is that, with RBF interpolants converging to polynomials, the boundary accuracy often deteriorates (the polynomial Runge phenomenon).

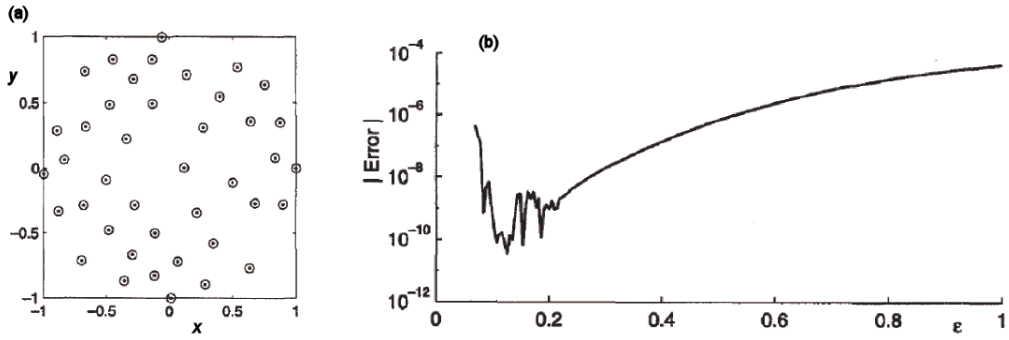


Figure 1: (a) A set of 41 scattered nodes in the unit circle, (b) The error in max norm when the test function $f(x,y) = 59/(67 + (x+\frac{1}{7})^2 + (y-\frac{1}{11})^2)$ is interpolated using these nodes, displayed as a function of the shape parameter ε. (Reproduced from [17].)

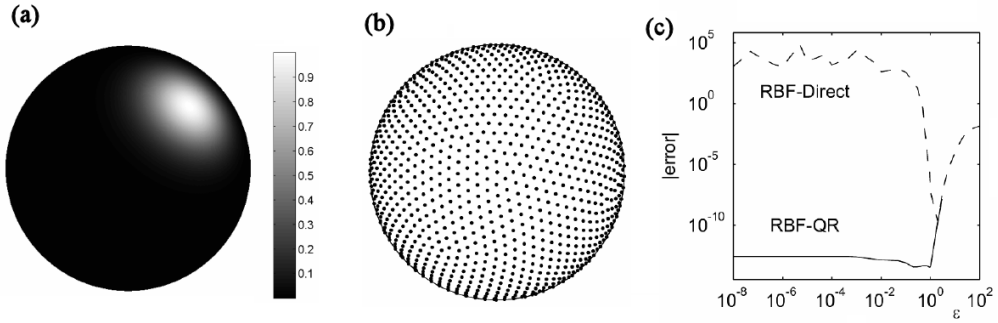


Figure 2: (a) Test function $f(x) = e^{-7(x+\frac{1}{2})^2 - 8(y+\frac{1}{2})^2 - 9(z-\frac{1}{\sqrt{2}})^2}$, (b) $n = 1849$ ME (minimal energy) nodes, (c) Interpolation errors (in max norm) when using RBF-Direct vs. using the stable RBF-QR algorithm.

2.2 The ill-conditioning of the A-matrix

Sideways translates of near-flat basis functions all look the same, and it is intuitively obvious that they must form a very ill-conditioned base to expand in. Just how bad it is can readily be quantified [18]. For example, for scattered nodes in 2-D, the eigenvalues of the A-matrix form distinct groups, following the specific pattern

$$\{O(1)\},\ \{O(\varepsilon^2), O(\varepsilon^2)\},\ \{O(\varepsilon^4), O(\varepsilon^4), O(\varepsilon^4)\},\ \{O(\varepsilon^6), O(\varepsilon^6), O(\varepsilon^6), O(\varepsilon^6)\},\ \ldots \tag{5}$$

until the last eigenvalue is reached (causing the last group to possibly contain fewer eigenvalues than the general pattern would suggest). Different choices of scattered node locations or of RBF

types (IQ, MQ, or GA), make no difference in this regard; lattice based nodes or Bessel-type RBFs form exceptions. More concisely, the eigenvalue pattern above can be written as

$$1, 2, 3, 4, 5, 6, \ldots , \tag{6}$$

indicating how many eigenvalues there are of orders ε^0, ε^2, ε^4, ε^6, ε^8, ε^{10}, etc. Table 2 shows some more such sequences. For d-D non-periodic domains, the sequence agrees with the dth diagonal in Pascal's triangle. Given these patterns, one can immediately calculate the orders of both cond(A) and $\det(A) = \prod_{k=1}^{n} \lambda_k$ as functions of n. For the examples in Figures 1 and 2, cond(A) becomes equal to $O(\varepsilon^{-16})$ and $O(\varepsilon^{-84})$, respectively.

Table 2: Numbers of eigenvalues of different sizes (powers of ε) for different geometries.

	Power of ε =								
Geometry	0	2	4	6	8	10	12	14	...
1-D non-periodic	1	1	1	1	1	1	1	1	...
1-D on circle periphery	1	2	2	2	2	2	2	2	...
2-D non-periodic	1	2	3	4	5	6	7	8	...
2-D on spherical surface	1	3	5	7	9	11	13	15	...
3-D non-periodic	1	3	6	10	15	21	28	36	...

2.3 Overview of some well-conditioned algorithms

In cases when RBF-Direct is too ill-conditioned, the most straightforward approach is to resort to extended precision arithmetic. The only drawback is that the cost usually becomes excessive. Given the results quoted in the previous section, one can readily determine in advance just how many digits of precision would be needed as function of n and ε in various geometrical settings.

Some types of preconditionings and SVD enhancements have been suggested for the RBF-Direct approach. While preconditioning can speed up certain iterative procedures, cf. Chapter 34 in [12], this does not address the issue that significant information becomes lost already when the coefficient matrix A is formed (with all its entries virtually the same when ε is small). Recovery of such missing information is challenging or impossible.

Stable algorithms use computational steps that remain numerically well conditioned all the way into the $\varepsilon \to 0$ limit (and therefore require only standard double precision arithmetic, no matter how small ε is). So far, two main classes of stable algorithms have been proposed. The first realizations of these were denoted *Contour-Padé* [17] and *RBF-QR* [16], respectively. Related to the latter is the recent *RBF-GA* algorithm [22].

2.3.1 The Contour-Padé algorithm

Although ε is a real-valued quantity, nothing stops us from considering it also for complex values. Focusing on the GA case, it can be shown that the interpolant $s(\mathbf{x}, \varepsilon)$, for any fixed evaluation point $\mathbf{x}$, then becomes a meromorphic function of ε (i.e. with poles as its only singularities across the finite complex ε-plane). Furthermore, it is known that $s(\mathbf{x}, 0)$ is finite even as $\varepsilon \to 0$. The origin $\varepsilon = 0$ must therefore be a removable singularity of $s(\mathbf{x}, \varepsilon)$, i.e. there is no actual singularity in the flat basis function limit as far as the RBF interpolant is concerned. The Contour-Padé algorithm is based on Cauchy's integral theorem, allowing the evaluation of an analytic function at a point (such as $\varepsilon = 0$) using an integration path that does not need to come close to it. The path can thus

follow such a large circle in the ε-plane that RBF-Direct can safely be used along it. The algorithm made possible the first demonstration that the flat limit of $\varepsilon \longrightarrow 0$ can be computationally highly attractive for the task of numerically solving PDEs [35].

2.3.2 The RBF-QR algorithm

As noted above, translates of near-flat RBFs form a basis that is ill-suited for immediate numerical use. This naturally raises the question whether the underlying approximation *space* also is bad, or if all conditioning issues can be resolved by just by finding an alternate good basis in exactly the same space. This latter case turns out to hold true, leading to the follow-up issue of how one can carry out the basis conversion by analytic means also in scattered node cases, i.e. so that no numerical cancelations will arise in the process. The RBF-QR method offers a systematic approach for this, first implemented for nodes on the surface of a sphere [16], and more recently for node sets in 1-D, 2-D and 3-D (however then limited to GA RBFs [21]) .

2.3.3 The RBF-GA algorithm

The RBF-QR algorithm involves extensive manipulations of Taylor expansions. While these can be truncated as needed, the RBF-GA algorithm utilizes that exact remainders can be expressed as incomplete gamma functions. This leads to a stable algorithm that is free from both infinite expansions and inexact truncations. It applies to GA RBFs in any number of dimensions, and is presently the fastest stable option available (at around 10 times the cost of RBF-Direct). Although it may be slightly less accurate than RBF-QR in some cases (such as for large lattice-like node sets), it is nevertheless well suited for generating RBF-FD approximations.

2.4 Time stabilization: hyperviscosity

When carrying out explicit time stepping of wave-type PDEs using RBF-FD-based spatial discretization, another type of stability issue emanates from the fact that the natural intrinsic irregularity of the RBF-FD stencils causes eigenvalues of the DM to scatter into the right half of the complex plane. This becomes a hurdle to the RBF-FD method especially when using large RBF-FD stencils since, as the stencil size increases, so does the scatter of eigenvalues. This latter point is even an issue for systems with dissipation, in which case the scatter might be too large for the natural dissipation to control.

Stabilization of the RBF-FD method is achieved by applying a hyperviscosity filter, that is a small constant times a high-order Laplacian $\Delta^k, k \in \mathbb{N} > 1$, to the right hand side (RHS) of the system of PDEs being solved. Since the stencils for the hyperviscosity are same size as those used in discretizing the spatial operators and are simply added to the RHS, there is no additional cost per time [13; 15]. If GA RBFs ($\phi(r) = e^{-(\varepsilon r)^2}$) are used for creating the hyperviscosity stencils, the values for $\Delta^k \phi(r)$ are available explicitly, thanks to the relation

$$\Delta^k \phi(r) = \varepsilon^{2k} p_k(r) \phi(r). \tag{7}$$

Here, k is the order of the Laplacian and $p_k(r)$ are multiples of generalized Laguerre polynomials that can be generated recursively (see Section 3.2 [15]). A 2D Laplacian operator is assumed when working on the surface of the sphere since a local stencil can be viewed as planar.

Applying hyperviscosity leaves the physically relevant eigenvalues largely intact but shifts all the spurious ones of the PDE system to the left half of the complex plane. This shift is controlled by k, the order of the Laplacian, and a scaling parameter γ_c, defined by

$$H = \gamma \Delta^k = \gamma_c N^{-k} \Delta^k.$$

where N is the total number of nodes in the domain. Given a choice of N, n, ε, it was found experimentally that γ_c ranging from $O(1)$ to $O(10^{-2})$ provides stability with good accuracy for PDEs with convective operators [13]. In general, the larger the stencil size, the higher the order of the Laplacian should be used to give the best results. This is attributed to the fact that as the stencil size n increases, the accuracy of the RBF-FD method increases; that is, a wider range of physical modes represented accurately. As a result, the hyperviscosity operator should preserve as much of that range as possible; as k increases, more physical modes are preserved correctly.

3 Implicit (compact) RBF-FD approximations

Since a derivative is a 'local' property of a function, there is something intuitively contradictory about enhancing the accuracy of a FD approximation by invoking data located increasingly far away. When the task is to solve a PDE (rather than just to approximate an operator), *compact approximations* offer a different opportunity for improving the order of accuracy. For finite differences, the concept has a long history [9; 23] with several more recent enhancements available (such as to nonlinear PDEs in 2-D and 3-D, etc.) [30; 36; 37; 64].

Before considering compact approximations in scattered node RBF-FD cases, the basic idea is illustrated in the case of approximating $\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ on a 2-D lattice, with spacing h in each direction. The most obvious FD approximation can be written as

$$\begin{bmatrix} & 1 & \\ 1 & -4 & 1 \\ & 1 & \end{bmatrix} u/h^2 = \Delta u + O(h^2)\,. \tag{8}$$

Using only a 3×3 stencil size, it is impossible to find weights that improve the accuracy above second order. Extending the stencil to 5 weights in both directions permits fourth order, but causes problems when solving the PDE $\Delta u = f$:

- The center weight becomes smaller in magnitude than the sum of magnitudes of the remaining weights, i.e. *diagonal dominance* is lost. This damages the convergence rate of many iterative schemes, and it also opens up the possibility of system singularities.
- Wider stencils need more boundary information than what is readily available.

Another way to approximate Δu is given by

$$\begin{bmatrix} 1 & 4 & 1 \\ 4 & -20 & 4 \\ 1 & 4 & 1 \end{bmatrix} u/(6h^2) = \Delta u + O(h^2)\,, \tag{9}$$

with no immediately obvious advantage over (8). However, Taylor expansions will reveal that, for solving $\Delta u = f$,

$$\begin{bmatrix} 1 & 4 & 1 \\ 4 & -20 & 4 \\ 1 & 4 & 1 \end{bmatrix} u/(6h^2) = \begin{bmatrix} & 1 & \\ 1 & 8 & 1 \\ & 1 & \end{bmatrix} f/12 + O(h^4),$$

and for solving $\Delta u = 0$,

$$\begin{bmatrix} 1 & 4 & 1 \\ 4 & -20 & 4 \\ 1 & 4 & 1 \end{bmatrix} u/(6h^2) = 0 + O(h^6)\,.$$

The latter two approximations suffer neither of the problems noted above, but achieve nevertheless significantly improved levels of accuracy.

The weights a_i in 'regular' RBF-FD approximations of a linear operator L at a node $\underline{x}_1$ are obtained by requiring that $Lu|_{\underline{x}=\underline{x}_1} = \sum_{i=1}^{n} a_i u_i$ is exact for all the RBFs $\phi(||\underline{x} - \underline{x}_i||)$, leading to an $n \times n$ linear system for the weights (system represented in (4) without column and row of ones). For compact RBF-FD approximations, first described in [60], we again consider a RBF-FD stencil centered at $\underline{x}_1$, with further nodes $\underline{x}_2, \ldots, \underline{x}_n$, and then repeat $m \leq n-1$ of the latter ones: $\underline{x}_{\sigma_1}, \ldots, \underline{x}_{\sigma_m}$. Wishing to solve $Lu = f$, we extend the basis function set to also include $L\phi(||\underline{x} - \underline{x}_{\sigma_j}||)$ and require $Lu|_{\underline{x}=\underline{x}_1} = \sum_{i=1}^{n} a_i u|_{\underline{x}=\underline{x}_i} + \sum_{j=1}^{m} b_j Lu|_{\underline{x}=\underline{x}_{\sigma_j}}$ to hold for them all. This leads to a $(n+m)\times(n+m)$ linear system for the weights a_i and b_j. With $Lu|_{\underline{x}=\underline{x}_1} = f_1$, $u|_{\underline{x}=\underline{x}_i} = u_i$ and $Lu|_{\underline{x}=\underline{x}_{\sigma_j}} = f_{\sigma_j}$, we have thus arrived at the desired compact RBF-FD formula (sometimes denoted RBF-HFD for being based on a Hermite type interpolation). The usual enhancement of also enforcing exact results for $u(\underline{x}) \equiv 1$ together with the constraint $\sum_{i=1}^{n} \lambda_i = 0$ is again beneficial for accuracy, and is readily incorporated. The resulting $(n+m+1) \times (n+m+1)$ linear system will again generally be positive definite.

The advantages noted above for compact formulas in the FD case carry directly over to the scattered node RBF-FD case. Several test examples and further discussions are provided in [60].

4 Applications on the surface of spheres and spheroids

4.1 Cartesian form of surface differential operators

Spherical coordinates are well-known to suffer from the "pole problem". When expressing PDEs in this coordinate system, this problem is exacerbated by the fact that directional velocity *vector* components in the lateral direction u (latitudinal) and v (longitudinal) will inherently carry pole singularities in their solution since the unit vectors $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\lambda}}$ are singular at the poles. Similar issues occur when representing more general surfaces in surface-based (or intrinsic) coordinate systems, such as oblate or prolate spheroidal coordinates. If, however, a 3-D Cartesian coordinate system is used, these singularities can be completely avoided. This section describes how to express common surface differential operators in Cartesian form. The discussion below is for general two-dimensional surfaces embedded in three dimensional space, with some specific comments on the sphere and spheroids toward the end. These Cartesian based operators are completely natural to apply to an RBF expansion since the radial functions are free of any surface based coordinate systems.

Let $\mathbf{x} = (x, y, z)$ be a point on the target surface and $\mathbf{n} = (n_x, n_y, n_z)$ denote the unit normal vector to the surface at $\mathbf{x}$. If $\mathbf{u} = (u, v, w)$ is a vector expressed in Cartesian coordinates and is placed at (x, y, z), then $\mathbf{n}(\mathbf{n} \cdot \mathbf{u}) = \mathbf{n}\mathbf{n}^T\mathbf{u}$ gives the projection of $\mathbf{u}$ onto the normal to the surface, and $\mathbf{u} - \mathbf{n}\mathbf{n}^T\mathbf{u}$ gives the projection of $\mathbf{u}$ onto the plane tangent to the surface at (x, y, z). This projection operation can be expressed using following projection matrix:

$$\mathbf{P} = \mathbf{I} - \mathbf{n}\mathbf{n}^T = \begin{bmatrix} (1-n_x n_x) & -n_x n_y & -n_x n_z \\ -n_x n_y & (1-n_y n_y) & -n_y n_z \\ -n_x n_z & -n_y n_z & (1-n_z n_z) \end{bmatrix} = \begin{bmatrix} \mathbf{p}_x^T \\ \mathbf{p}_y^T \\ \mathbf{p}_z^T \end{bmatrix}, \tag{10}$$

where $\mathbf{I}$ is the 3-by-3 identity matrix. Here $\mathbf{p}_x$, $\mathbf{p}_y$, $\mathbf{p}_z$ represent the projection operators in the x, y and z directions, respectively. This projection operator can be combined with the standard Cartesain-based gradient operator ∇ to define a variety of surface differential operators in Cartesian coordinates. Table 3 summarizes these results.

Table 3: Common surface differential operators expressed in Cartesian coordinates. The projection matrix $\mathbf{P}$ is defined in (10) and ∇ is the standard Cartesian based gradient, i.e., $\nabla = \partial_x\hat{\mathbf{i}} + \partial_y\hat{\mathbf{j}} + \partial_z\hat{\mathbf{k}}$.

Surface differential operator	Expression in Cartesian coordinates
Gradient of a scalar h	$\mathbf{P}\nabla h = \begin{bmatrix} (\mathbf{p}_x \cdot \nabla)h \\ (\mathbf{p}_y \cdot \nabla)h \\ (\mathbf{p}_z \cdot \nabla)h \end{bmatrix}$
Divergence of a vector $\mathbf{u} = (u, v, w)$	$(\mathbf{P}\nabla) \cdot \mathbf{u} = (\mathbf{p}_x \cdot \nabla)u + (\mathbf{p}_y \cdot \nabla)v + (\mathbf{p}_z \cdot \nabla)w$
Curl of a vector $\mathbf{u} = (u, v, w)$	$(\mathbf{P}\nabla) \times \mathbf{u} = \begin{bmatrix} (\mathbf{p}_y \cdot \nabla)w - (\mathbf{p}_z \cdot \nabla)v \\ (\mathbf{p}_z \cdot \nabla)u - (\mathbf{p}_x \cdot \nabla)w \\ (\mathbf{p}_x \cdot \nabla)v - (\mathbf{p}_y \cdot \nabla)u \end{bmatrix}$
Laplace-Beltrami of scalar h	$(\mathbf{P}\nabla) \cdot (\mathbf{P}\nabla h) = (\mathbf{p}_x \cdot \nabla)(\mathbf{p}_x \cdot \nabla)h + (\mathbf{p}_y \cdot \nabla)(\mathbf{p}_y \cdot \nabla)h + (\mathbf{p}_z \cdot \nabla)(\mathbf{p}_z \cdot \nabla)h$

In the case of the unit sphere, the normal vector $\mathbf{n}$ at a point $\mathbf{x} = (x, y, z)$ on the sphere is just $\mathbf{x}$. In this case, the projection operator simplifies to

$$\mathbf{P} = \mathbf{I} - \mathbf{x}\mathbf{x}^T = \begin{bmatrix} (1-x^2) & -xy & -xz \\ -xy & (1-y^2) & -yz \\ -xz & -yz & (1-z^2) \end{bmatrix}. \tag{11}$$

For a spheroid defined by

$$\frac{x^2 + y^2}{a^2} + \frac{z^2}{c^2} = 1, \tag{12}$$

the unit normal vectors are given by

$$\mathbf{n} = (n_x, n_y, n_z) = \left(\frac{2x}{a^2\xi}, \frac{2y}{a^2\xi}, \frac{2z}{c^2\xi} \right), \tag{13}$$

where $\xi = \sqrt{(x^2 + y^2)/a^4 + z^2/c^4}$.

4.2 Non-linear shallow water equations on a sphere

The shallow water equations describe the nonlinear flow of an incompressible fluid. They represent the 2-D atmospheric flow conditions in a single hydrostatic atmospheric layer and are therefore considered an idealized test-bed for the horizontal dynamics (known as the dynamical core) of all 3-D climate model developments. The equations address the majority of the modeling challenges associated with the temporal and horizontal discretization techniques in spherical geometry.

Using the notation from Table 3, the shallow water equations on the surface of the unit sphere

in Cartesian coordinates are given by

$$\frac{\partial u}{\partial t} = -\,\mathbf{p_x}\cdot\underbrace{\begin{bmatrix}(\mathbf{u}\cdot\mathbf{P}\nabla)u + f(\mathbf{x}\times\mathbf{u})\cdot\hat{\mathbf{i}} + g(\mathbf{p}_x\cdot\nabla)h\\ (\mathbf{u}\cdot\mathbf{P}\nabla)v + f(\mathbf{x}\times\mathbf{u})\cdot\hat{\mathbf{j}} + g(\mathbf{p}_y\cdot\nabla)h\\ (\mathbf{u}\cdot\mathbf{P}\nabla)w + f(\mathbf{x}\times\mathbf{u})\cdot\hat{\mathbf{k}} + g(\mathbf{p}_z\cdot\nabla)h\end{bmatrix}}_{\text{RHS}}, \tag{14a}$$

$$\frac{\partial v}{\partial t} = -\,\mathbf{p}_y\cdot\text{RHS}, \tag{14b}$$

$$\frac{\partial w}{\partial t} = -\,\mathbf{p}_z\cdot\text{RHS}, \tag{14c}$$

$$\frac{\partial h}{\partial t} = -\,(\mathbf{P}\nabla)\cdot(h\mathbf{u})\,. \tag{14d}$$

where $\mathbf{P}$ is the projection operator, given in (11), and confines the flow to the sphere, f is the Coriolis force, $\mathbf{u} = u\hat{\mathbf{i}} + v\hat{\mathbf{j}} + w\hat{\mathbf{k}}$ is the velocity vector, $\mathbf{x}$ represents the position vector, and h is the geopotential height. Notice that the only spatial operators that need to be discretized are the components of the surface gradient $(\mathbf{p}_x\cdot\nabla), (\mathbf{p}_y\cdot\nabla), (\mathbf{p}_z\cdot\nabla)$. These operators are used as L in (4) to generate RBF-FD approximations in the numerical test cases that follow.

4.2.1 Flow over an isolated mountain

This test case describes flow over a single isolated mountain, which is achieved by adding a forcing term h_{mtn} to the right hand side of the geopotential height h equation in (14d). The standard undifferentiable C^0 mountain is given by

$$h_{mtn} = h_{max}(1 - d/R) \tag{15}$$

where $h_{max} = 2000\,m$, $R = \pi/9$, and $d^2 = \min[R^2, (\lambda-\lambda_c)^2 + (\theta-\theta_c)^2]$ with ($\theta_c = 30^o$N, $\lambda_c = -90^o$W) being the center of the mountain. To differentiate between errors due to a non-smooth forcing, which causes Gibbs phenomena in any high-order method, and those inherent in the RBF-FD method, the results for convergence and accuracy are compared against test runs that use an exceptionally steep C^∞ Gaussian profile given by

$$h(\lambda,\theta) = h_{max}e^{-(2.8\frac{d}{R})^2}, \tag{16}$$

where h_{max}, d and R are the same as for the cone mountain. The initial conditions are given by

$$h = h_0 - \frac{1}{g}(a\Omega u_0 + \frac{u_0^2}{2})z^2, \qquad \mathbf{u} = u_0\{-y, x, 0\}, \tag{17}$$

where $h_0 = 5400\,m$ (mean reference height), $g = 9.80616\,\text{m}$, $u_0 = 20\,\text{m/s}$, $a = 6,371,220\,\text{m}$ (mean radius of the earth), and $\Omega = 7.292(10)^{-5}\text{s}^{-1}$ (rotation rate of the earth). The simulation is run for 15 days using the standard RK4 time-stepping scheme.

The left column of Figure 3 shows the profile of the conical mountain, the solution in h at Day 15 and the magnitude of the error between the RBF-FD solution for $N = 25,600$, $n = 31$ and the discontinuous Galerkin (DG) reference solution (see description below). The same holds for the right column but with the Gaussian mountain profile. With the solutions looking identically the same, the key difference to notice is that even though the C^∞ Gaussian mountain is slightly steeper than the C^0 mountain, no Gibbs phenomena is observed in the former. With the C^0 mountain, there are high frequency waves emanating throughout the domain (i.e. Gibbs' phenomena) , illustrating the

sensitivity of high-order methods to non-smooth forcing. Another consequence is that the accuracy of the RBF-FD method does not indefinitely increase with stencil size n, as shown in the bottom left panel of Figure 3. After $n = 31$, stencil size has no bearing on accuracy when a non-smooth mountain forcing is present. Hence, this is the stencil size chosen for this test case. In contrast, the bottom right panel of Figure 3 demonstrates how the accuracy of the RBF-FD method for a C^∞ solution does increase as n increases, that is as the derivative approximations become more global. However, even with a smooth forcing, the rate of convergence is not much greater than for the cone case, since both the Gaussian or cone mountains are so steep, leading to under-resolution even for very large nodes sets. To overcome this, adaptive node refinement in the area of the mountain needs to be used, as was done in [49].

Next the performance of RBF-FD method is compared not only against itself but also against a discontinuous Galerkin (DG) and spherical harmonic (SH) models.

1. DG: The DG model [4] is in 3-D Cartesian coordinates, with flows being tangentially constrained to the sphere by adding a Lagrange multiplier to the system of equations. The simulations used as references herein have been performed on a cubed sphere grid made up of 6144 elements. Each element contains 12x12 Legendre quadrature nodes to represent the solution, which results in a total of $884,736$ degrees of freedom and an average resolution around 26 km. For computing these reference solutions, no dissipation mechanism was found to be needed. However, for the run time versus error computations in Figure 5, the two dimensional exponential filter described in [33] was applied.

2. SH: The SH model from the DWD (Deutscher Wetterdienst, German National Weather Service, see `http://icon.enes.org/`) is an updated derivative of the NCAR spectral transform model. It is implemented with de-aliasing, using Orszag's 2/3 rule [29] and has become the standard reference solution in the community. For the flow over a mountain test, it has a spectral truncation of T426, that is it uses 182,329 spherical harmonic bases.

3. RBF-FD: A high resolution RBF-FD model is also used as a reference, based on $N = 163,824$ icosahedral-type nodes on the sphere, representing a 60km resolution. It uses a stencil size of $n = 31$.

The left panel of Figure 4 shows the normalized ℓ_2 error is an order of magnitude larger when the DWD-SH reference solution is used, as opposed to DG or the RBF-FD ($N = 163,842$) reference solutions. Furthermore when these latter reference solutions are used, the normalized ℓ_2 errors are almost identical (notice the $\bigcirc$ overlay the $\square$). This same trend is also seen in the right panel of Figure 4 with global RBFs, a different approach than both RBF-FD and DG that does not require hyperviscosity . Given that DG, RBF-FD, and global RBFs are vastly different numerical methods, strongly indicates that the DWD-SH T426 spectral simulation is providing a less accurate solution. This is further supported by the few articles that do report ℓ_2 errors for this test case [48; 49; 51], all of which use either the NCAR or DWD SH reference solution, and obtain errors on the order of 10^{-4}, an order of magnitude larger than that obtained by DG or RBF-FD.

Next, time benchmarking is considered with comparison against the DG model. Benchmarking was done on a MacBook Pro laptop with an Intel i7 2.2 GHz quad-core processor, using only a single core and 8 GB of memory. The RBF-FD code was written in MATLAB and the DG code in C++. The RBF-FD reference solution of $N = 163,842$ and $n = 31$ (i.e. 60 km resolution) was used for calculating the ℓ_2 error versus runtime (i.e. wall-clock time) for both methods in Figure 5. The RBF-FD resolutions with corresponding time steps is given in Table 4. The RBF-FD method was computationally faster than the DG method, from about an order and a half of magnitude for coarser resolutions to 4 times faster for the finest resolutions.

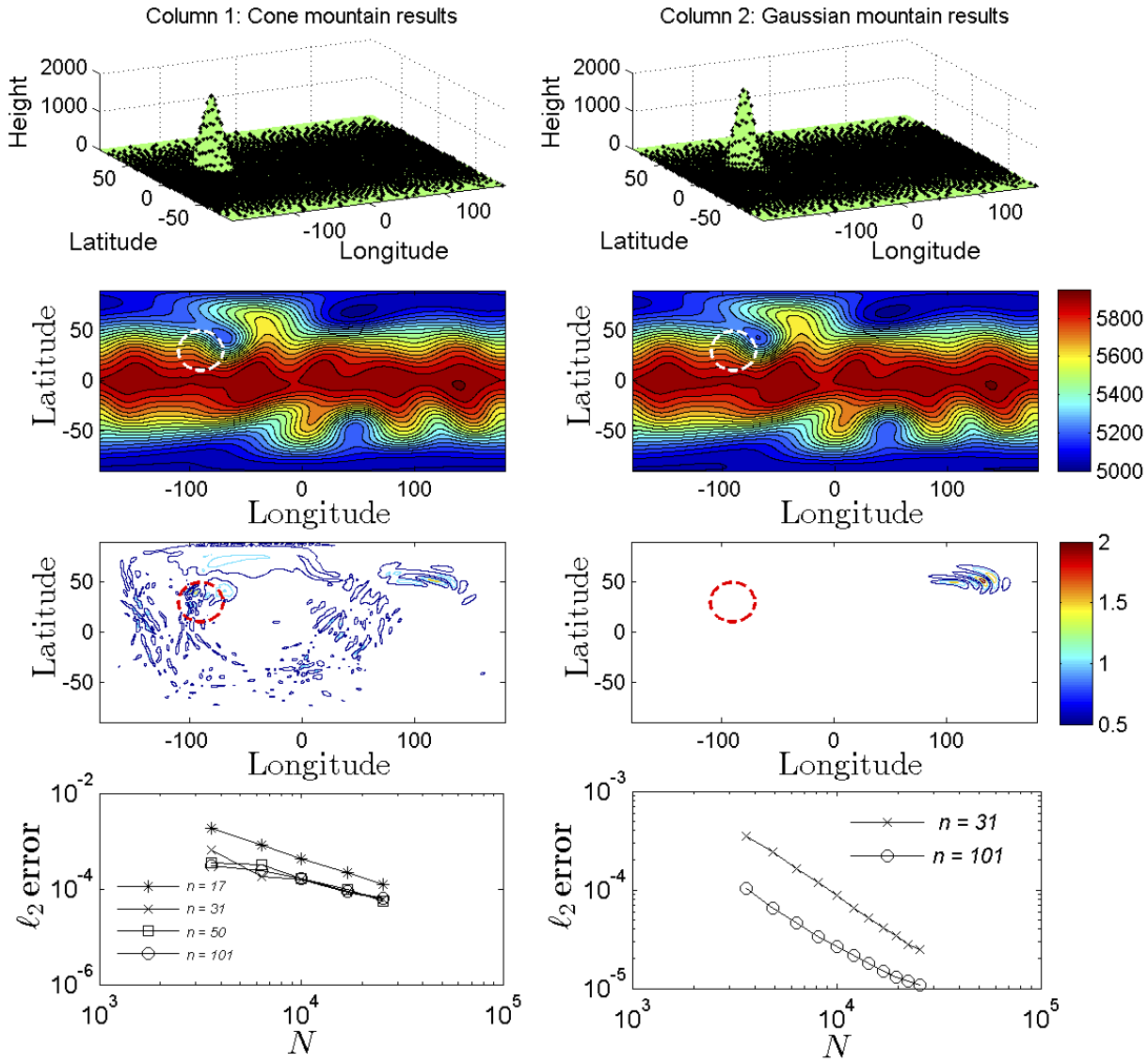


Figure 3: Left column- Cone mountain results: (1) profile of mountain; (2) RBF-FD solution for h at day 15, $N = 25,600$ and $n = 31$ with contour intervals at 50 m; (3) magnitude in the error between the RBF-FD solution and DG reference solution; (4) ℓ_2 error as function of the resolution N for varying stencil sizes. Right column, same as left but for the Gaussian mountain forcing. Dashed circle in all plots is the base of the mountain.

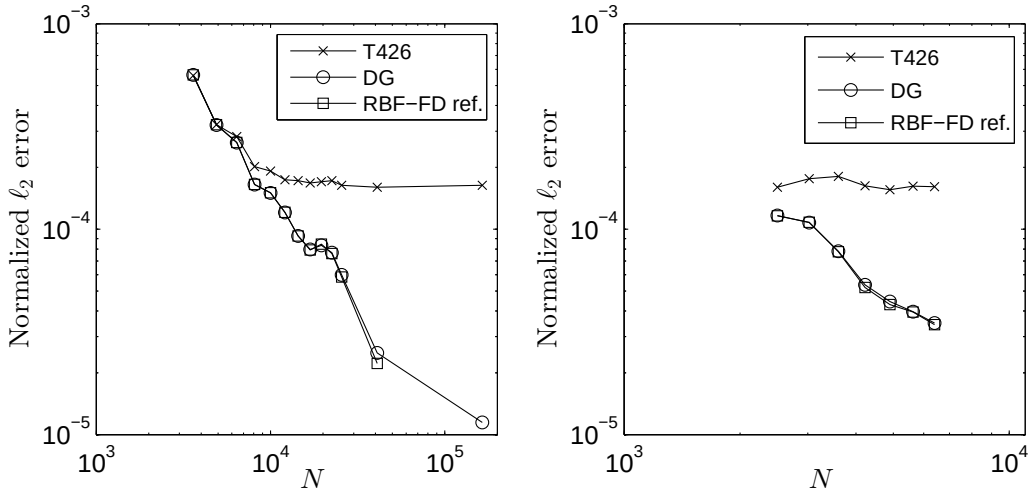


Figure 4: The normalized ℓ_2 error in the height field h as a function of N for flow over a conical mountain at day 15 for RBF-FD (left panel) and global RBFs (right panel). The different markers correspond to different reference solutions.

Table 4: Time steps used for the cone mountain case with respective spatial resolutions for the maximal determinant (MD) nodes based on the mesh norm, $\max_{\mathbf{x}\in\mathbb{S}^2} \min_{1\leq i\leq N} \text{dist}(\mathbf{x}, \mathbf{x}_i)$ (see [59] for discussion and tabulation) .

N	Resolution (km)	Node type	Δt (minutes)
4096	550	MD	20
6400	420		15
12100	330		12
25600	220		5
40962	120	ICO	3
163842	60		1

4.2.2 Evolution of a highly nonlinear wave

This test case describes the evolution of a highly nonlinear wave with rapid energy transfer from large to small scales over a short time period. First, high frequency gravity waves propagate around the sphere followed by complex vortex formations with sharp gradients. The details of how to set up the test can be found in [13; 28]. The background flow is only a function of latitude, represented by an exponential profile that is zero everywhere except in the latitudinal band $\pi/7 \leq \theta \leq 5\pi/14$. To generate the instability, the height (pressure) field is perturbed by Gaussians, in longitude and latitude, multiplied by a cosine to force the perturbation to go to zero at the poles $\theta = \pm\pi/2$. The test case is run for 6 days.

There are two main concerns with this test case: a) how well the sharp gradients are resolved and b) the effect of Gibbs phenomena. For *short* time integration periods, as here, these two numerical issues become a balancing act. As both the solution N and order of the method n are increased, the gradients are resolved better. However, as with classical FD, the higher the order of the method, i.e. larger n, the more prominent the Gibbs phenomena. Here with $N = 25,000$, $n = 101$ would correspond approximately to a ninth-order method. Notice in Figure 4.2.2, when an $n = 101$ stencil size is used, the contour lines are more jagged with evidence of of Gibbs phenomena.

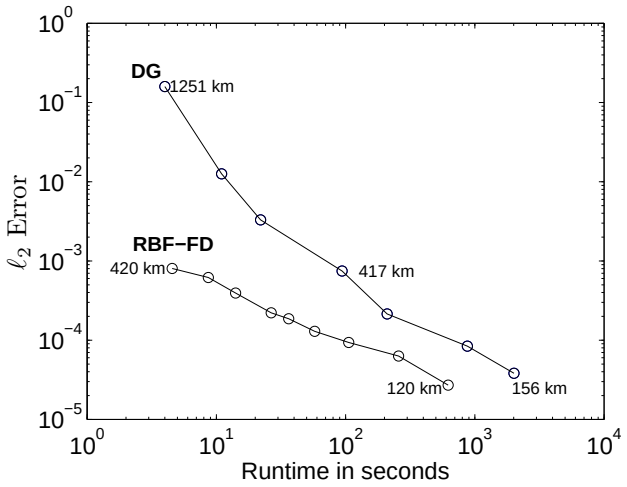


Figure 5: The error as a function of runtime (defined by wall clock time) for the flow over the cone mountain test case.

Yet, the gradients and vortices are much better defined than for $n = 31$, roughly analogous to a fifth-sixth order method. It this latter case, the features appear smoothed out and under-developed. Increasing the resolution N and keeping $n = 31$ fixed gives the best results as seen in Figure 7 for $N = 163,842$ (60km or $0.54^\circ \times 0.54^\circ$). The solution is extremely similar to that given by the high-order DG solution with a resolution of 39km or $0.35^\circ \times 0.35^\circ$. Due to its high accuracy in approximating derivatives, the RBF-FD method is able to produce the basic wave pattern structure even at very coarse resolutions such as $N = 4096$ ($5^\circ \times 5^\circ$), displayed in Figure 8. This is not the case with the DG method (a slightly higher resolution of $3^\circ \times 3^\circ$ is displayed in the third panel of Figure 8), a spectral element method (SE), or a finite volume method (FV). At such coarse resolutions, the DG and SE [49] methods instead produce features of the grid such as an artificial wavenumber four pattern for the cubed-sphere. The FV method, notorious for being dissipative, shows no spatial structures at a resolution of $5^\circ \times 5^\circ$ in Figure 8.

4.3 Reaction-diffusion equations on spheroids

While the sphere plays a prominent role in several applications from geoscience, it is often a physical idealization. In some cases, spheroidal domains (oblate or prolate) may be more physically relevant. Unfortunately, many numerical techniques that work well for the sphere, can be considerably more difficult in spheroidal domains (e.g., going from spherical harmonics to oblate/prolate spheroidal harmonics). Since RBFs are free of any surface based coordinate system, they do not suffer from these complications. This fundamental feature is illustrated in this section by considering the problem of simulating certain reaction-diffusion equations on the sphere, an oblate spheroid, and a prolate spheroid using the RBF-FD method . These types of equations and these domains have a number of applications in biology and chemistry to model such things as diffusion of chemicals on biological cells or membranes, pattern formations in biology, and nonlinear chemical oscillators in excitable media. While the discussion below is for the surface of a spheroid defined by (12), it may be possible to generalize it to more general (closed) surfaces, as has been done for the global RBF method in [27] and [42].

In the case of two species u and v, the prototypical form of reaction-diffusion equations on a

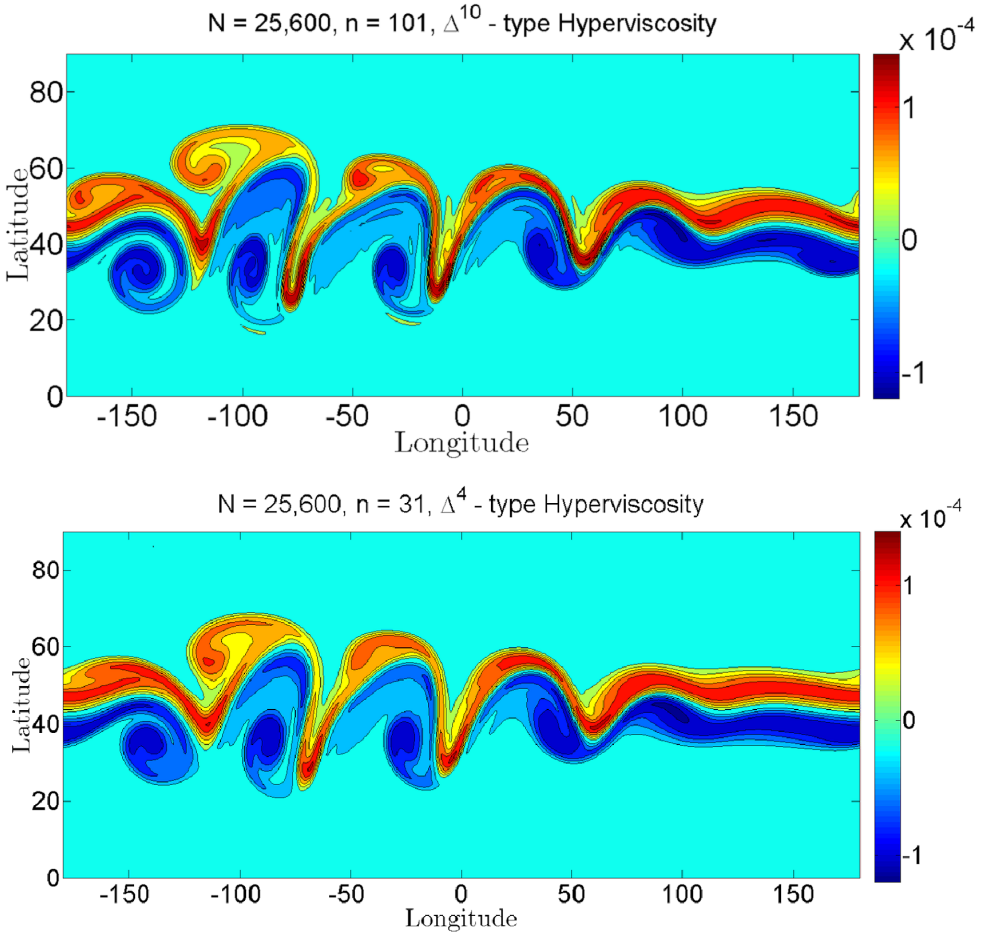


Figure 6: The relative vorticity at day 6 for the evolution of the highly unstable wave case as function of resolution N and stencil size n. Contour interval is $2 \cdot 10^{-5} s^{-1}$.

spheroid (or more general surface) can be written as

$$\begin{aligned} \frac{\partial u}{\partial t} =& \delta_u \Delta_{\mathcal{S}} u + f_u(t,u,v), \\ \frac{\partial v}{\partial t} =& \delta_v \Delta_{\mathcal{S}} v + f_v(t,u,v), \end{aligned} \tag{18}$$

where $\delta_u, \delta_v \geq 0$, f_u, f_v are typically non-linear scalar functions describing reactions of the species, and $\Delta_{\mathcal{S}}$ is the Laplace-Beltrami operator for the spheroid (or surface). As mentioned above, this operator can be difficult to treat numerically if using surface-based coordinate systems (i.e., oblate/prolate spheroidal coordinates) as they will induce an artificial coordinate singularity (e.g., at the poles). As in the case of the shallow water wave equations, Cartesian coordinates will thus be used to avoid these singularities. For (18), this means expressing $\Delta_{\mathcal{S}}$ using the formulation given in the last row of Table 3.

In the case of a sphere, the direct discretization of $\Delta_{\mathcal{S}}$ using RBFs is incredibly simple (as first discussed in [61] and employed below in Section 5.2). Unfortunately when switching to a more general spheroid, this simplicity is lost. A local version of the discretization approach taken in [27] is therefore adopted here. In this approach, N "scattered" nodes are first distributed on the surface of the spheroid. Each component of the surface gradient listed in top row of Table

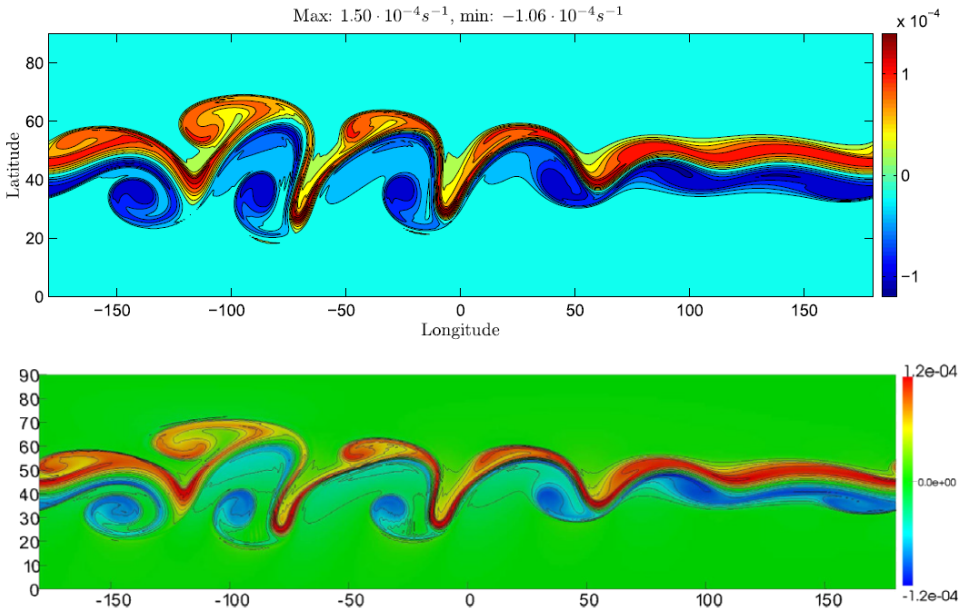


Figure 7: Top: RBF-FD solution for $N = 163,842$ and $n = 31$, Bottom: DG solution at 39 km resolution.

3 is then approximated with the RBF-FD method using the projection matrix $\mathbf{P}$ defined by the unit normal vectors in (13). A local approximation to the surface Laplacian is then constructed like the continuous formulation listed in last row of Table 3, but using the local RBF-FD DMs for each node. The weights for the local approximate surface Laplacian about the center node of the RBF-FD formula are then extracted and assembled into the global sparse DM for the approximate surface Laplacian.

Based on numerical experiments using the maximal determinant (MD) nodes [59] for the sphere radially projected to the spheroid, one small modification to the RBF-FD method described in Section 2 is suggested for the above procedure. Instead of computing the distance from node $\mathbf{x}_j$ to $\mathbf{x}_k$ using the standard Euclidean distance, a weighted Euclidean distance defined as

$$\|\mathbf{x}_j - \mathbf{x}_k\|_{\mathcal{S}}^2 = \frac{(x_j - x_k)^2 + (y_j - y_k)^2}{a^2} + \frac{(z_j - z_k)^2}{c^2}$$

should be used, where a and c define the parameters of the spheroid in (12). Although the results are not presented here due to space limitations, this was observed to give more accurate and stable approximations of the surface Laplacian. It should be noted that this modification does not alter the non-singularity results of the linear systems involved in computing the RBF-FD formulas discussed in Section 2.

4.3.1 Turing patterns

Since Turing's classical paper [53] that suggested how certain non-linear models of reaction and diffusion can lead to stable, heterogeneous pattern formations, there has been an explosion of research in reaction-diffusion-type models for various kinds of morphogenesis. The following is one

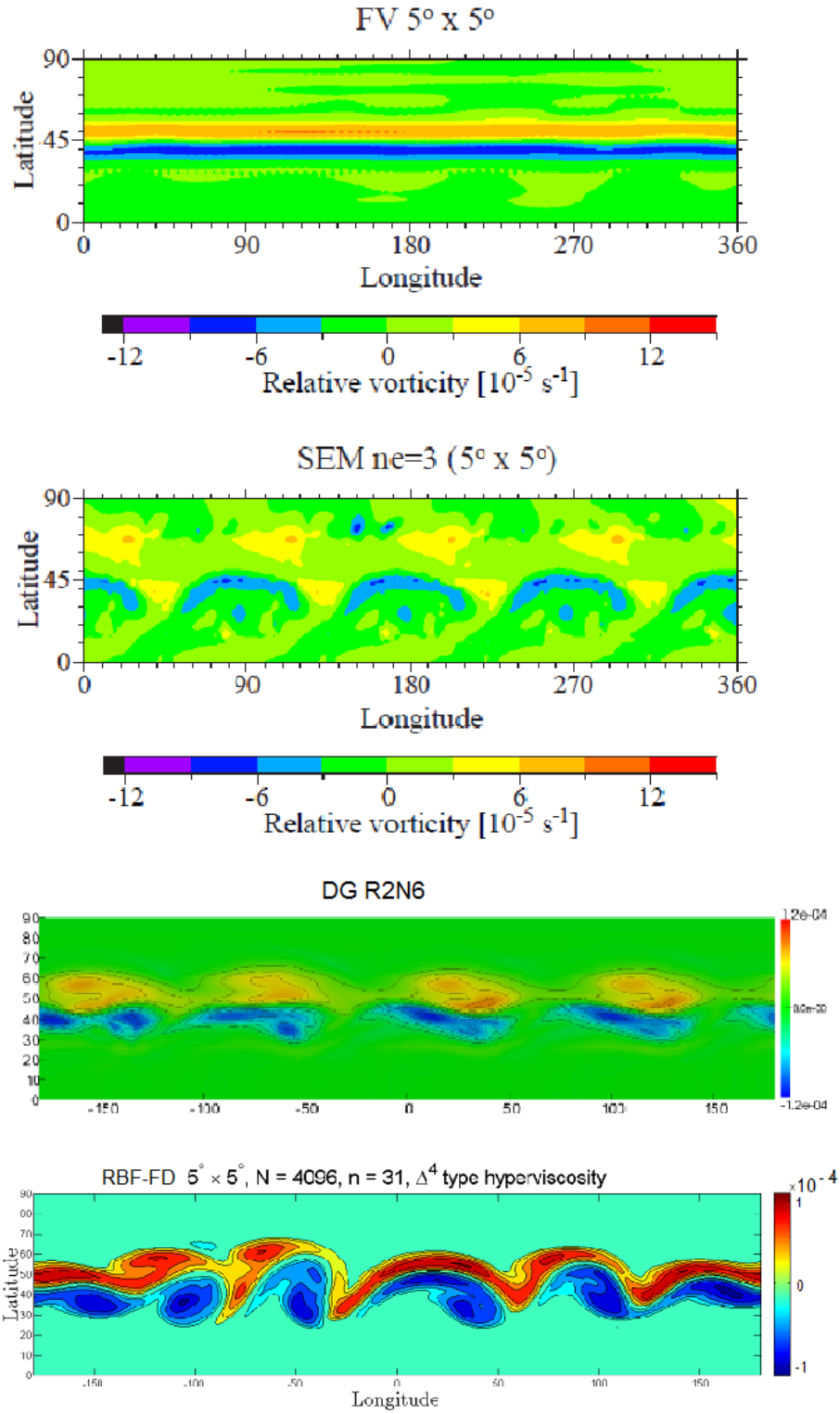


Figure 8: The relative vorticity at day 6 for the evolution of the highy unstable wave case with resolution of $5° \times 5°$ for a spectral element model (SEM), finite volume model (FV) [49], and the RBF-FD model. The DG model is approximately a resolution of $3° \times 3°$. R2 corresponds to 16 elements per face of the cubed sphere for a total of 96 elements, while N6 corresponds using $6 \times 6 = 36$ Legendre quadrature nodes. In the SEM model, $ne = 3$, corresponds to $3 \times 3 = 9$ elements per face of the cubed sphere. In each element 8×8 Gauss-Legendre-Lobatto nodes are used.

such model that has been studied on spherical domains [54]:

$$\begin{aligned}\frac{\partial u}{\partial t} =&\delta_u \Delta_{\mathcal{S}} u + \alpha u(1-\tau_1 v^2) + v(1-\tau_2 u),\\ \frac{\partial v}{\partial t} =&\delta_v \Delta_{\mathcal{S}} v + \beta v\left(1+\frac{\alpha\tau_1}{\beta}uv\right) + u(\gamma+\tau_2 v).\end{aligned} \tag{19}$$

Here u and v are morphogens with u the "activator" and v is the "inhibitor". If $\alpha = -\gamma$ then $(u, v) = (0, 0)$ is a unique equilibrium point of this system. By changing the diffusivity rates of u and v an instability can form that leads to different pattern formations. The cubic coupling parameter τ_1 favors the formation of stripes, while the quadratic coupling parameter τ_2 favors the formation spots. The spot pattern formations are more robust than stripes and take far less time to reach "steady-state".

The RBF-FD method was applied to numerically solving (19) on three surfaces: an oblate spheroid defined by (12) with $a = 1$ and $c = 0.5$, a prolate spheroid with $a = 1$ and $c = 1.5$, and the unit sphere (i.e. $a = c = 1$). The surfaces were discretized using a radial projection of a set of N spherical MD nodes to the surface. In all experiments, $N = 16,384$ and $n = 31$ for the RBF-FD approximations of the surface Laplacian. The equations were advanced in time using the third-order, semi-implicit, backward differentiation formulae (SBDF3) method [1]. This scheme treats the diffusion terms implicitly and the (non-linear) reaction terms explicitly. The implicit systems were solved using the (unpreconditioned) BiCGSTAB iterative method [55] with a tolerance of 10^{-8} on the relative residual. A time-step of $\Delta t = 0.05$ was used, which was near the maximum that could be used and still maintain time-stability due to the non-linear reaction terms. Throughout the simulations, the largest number of BiCGSTAB iterations required to solve the implicit systems was 5. Similar to the experiments in [54], the initial values of u and v were set to random values between -0.5 and 0.5 in a thin strip around the "equator" of each surface and $u = v = 0$ elsewhere.

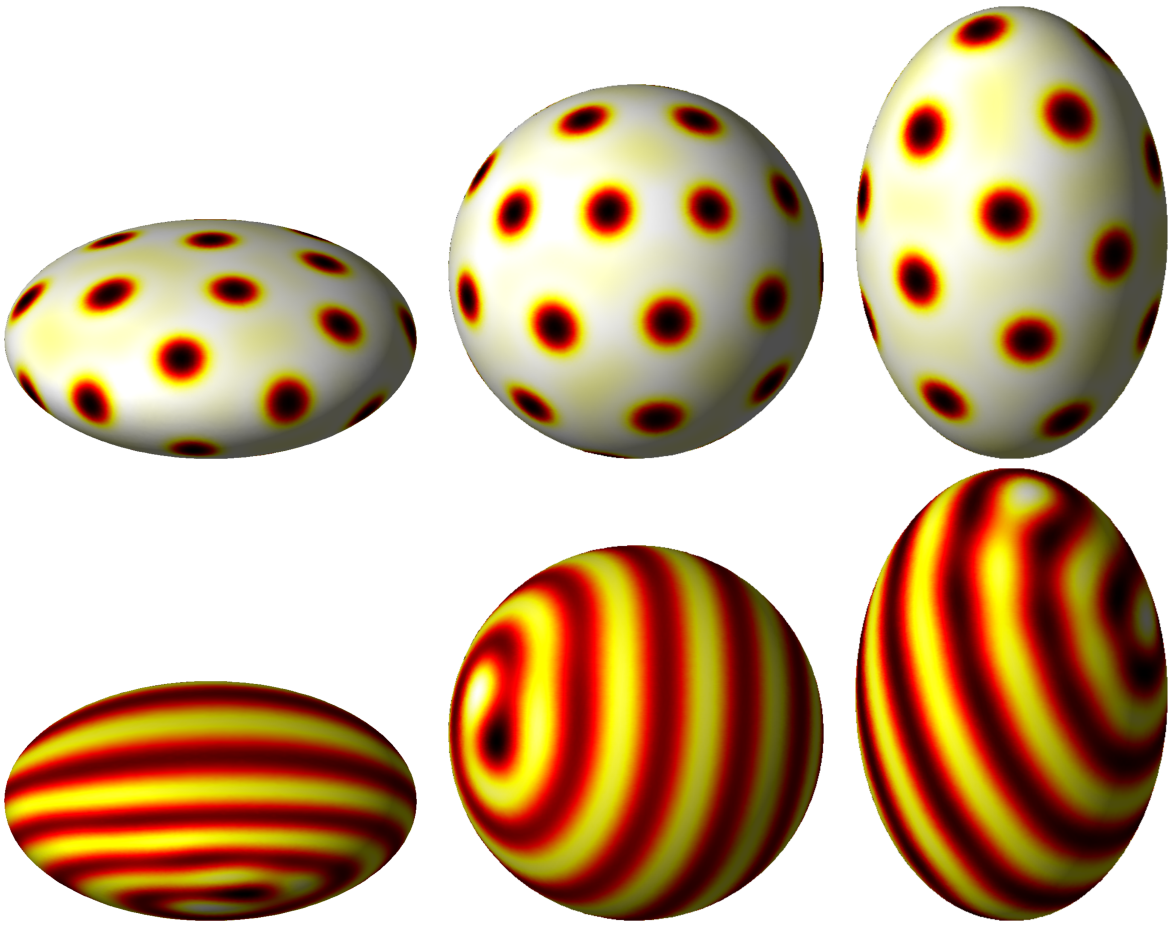
Figure 9 shows the results from the simulations using parameters that lead to both spot and stripe patterns. These parameter values were motivated by those used in [54]. For the spot patterns, the parameters $\alpha = 0.899$, $\beta = -0.91$, $\gamma = -0.899$, $\tau_1 = 0.02$, and $\tau_2 = 0.2$ were used for all the surfaces. The diffusion values were set as $\delta_u = 0.516\delta_v$, with $\delta_v = 2.25 \cdot 10^{-3}$ for the oblate spheroid, for $\delta_v = 3 \cdot 10^{-3}$ for the sphere, and $\delta_v = 4.5 \cdot 10^{-3}$ for the prolate spheroid. For the stripe patterns, $\tau_1 = 3.5$ and $\tau_2 = 0$, with α, β, and γ set the same as the spots for all the surfaces. The diffusion values were set as $\delta_u = 0.516\delta_v$, with $\delta_v = 1.575 \cdot 10^{-3}$ for the oblate spheroid, for $\delta_v = 2.1 \cdot 10^{-3}$ for the sphere, and $\delta_v = 3.15 \cdot 10^{-3}$ for the prolate spheroid.

While the initial conditions used in these simulations are random, the resulting spot and stripe patterns observed in Figure 9 are qualitatively similar to those obtained from the global RBF-method from [27]. This new RBF-FD method is significantly more computationally efficient since iterative methods can be used to successfully solve the implicit linear systems from the time-stepping scheme as opposed to direct methods used in [27]. This also allows much higher resolutions to be used in the simulations.

5 3-D applications in a spherical shell

5.1 Global electric circuit

Electrical linkages within the atmosphere are often discussed in terms of a "global electric circuit" (GEC). The GEC extends from the Earth's surface to the base of the ionosphere defined to be 90 km altitude. The basic idea, first postulated by Wilson [57; 58], is that thunderclouds and other highly electrified clouds produce an upward current ($\sim 1000 - 2000$ A) that maintains the ionosphere at



(a) (b) (c)

Figure 9: Turing spot and stripe patterns computed from the model (19) on an (a) oblate spheroid, (b) sphere, and (c) prolate spheroid). The pseudocolor plots are for the activator u once steady state is reached. In all plots, black corresponds to a high concentration of u, and white to a low concentration. For all simulations, $N = 16,384$ and $n = 31$, with a time-step of $\Delta t = 0.05$. All simulations were run until a steady-state solution was reached, which was about 800 time-steps for the spot patterns and 6000 for the strip patterns.

a quasi-static potential on the order 240 ± 40 kV with respect to the ground. A downward return current density of 1-10 $p\text{Am}^{-2}$ is distributed over the rest of the globe in so-called "fair weather regions" (no cloud activity).

The quasi-static coupled atmospheric and ionospheric components of the GEC can be modeled as an elliptical Poisson type PDE

$$\nabla \cdot (\sigma \nabla \Phi) = J_s(\vec{x}), \quad \vec{x} \in \Omega, \tag{20}$$

with *suitable boundary conditions*. Here, Φ is the electrostatic potential, Ω is the atmosphere below the ionosphere, and σ is the *atmospheric conductivity*. The right-hand side, J_s, is a representation of the *current sources* (e.g., thunderstorms) in the domain. The conductivity of the atmosphere is σ and to a good first approximation it can be considered simply as a function of altitude. Below, it is even further simplified by taking it to be a constant, thus resulting in Poisson's equation. This allows for a simple direct comparison between explicit RBF-FD stencil implementation to an

implicit (compact) RBF-FD one in 3-D spherical geometry, with the focus being on accuracy and convergence of a Krylov iterative solver (BiCGSTAB [55] in this case).

5.1.1 Poisson's equation

Consider Poisson's equation with homogeneous Dirichlet conditions, given by

$$\Delta u = f, \qquad \mathbf{x} \in \Omega, \tag{21}$$

$$u = 0, \qquad \mathbf{x} \in \partial\Omega, \tag{22}$$

where Ω is a spherical shell with inner radius $R_\mathrm{i} = 2$ and outer radius $R_\mathrm{o} = 4$. The right-hand side is chosen such that the exact solution is given by

$$u_{\text{exact}}(r, \theta, \varphi) = \sin\left(2\pi \frac{r - r_i}{r_0 - r_i}\right) Y_{9,3}(\theta, \varphi), \tag{23}$$

where $Y_{l,m}(\theta, \varphi)$ denotes a spherical harmonic of degree l and order m.

Two discretizations are considered, based on RBF-FD. The first is an explicit scheme, where the angular terms of the Laplacian in (21) are implemented using 31 point RBF-FD stencils on scattered nodes, and the radial terms are discretized using 5 point RBF-FD stencils on equidistant nodes. The second scheme is implicit, based on the methodology in [60] and discussed in Section 3. For the angular terms in the the Laplacian, $\{n = 9, m = 6\}$ node stencils are applied, while $\{n = 3, m = 2\}$ nodes are used in the radial discretization. Maximum determinant nodes (MD), see [59], are used in the angular discretization for both stencil types. A typical stencil of each kind is shown in Figure 10. The sparsity pattern of the Laplacian DM for each method, following a

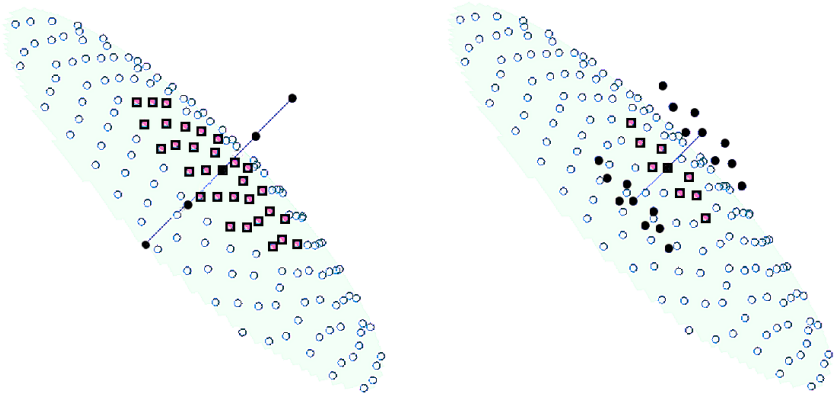
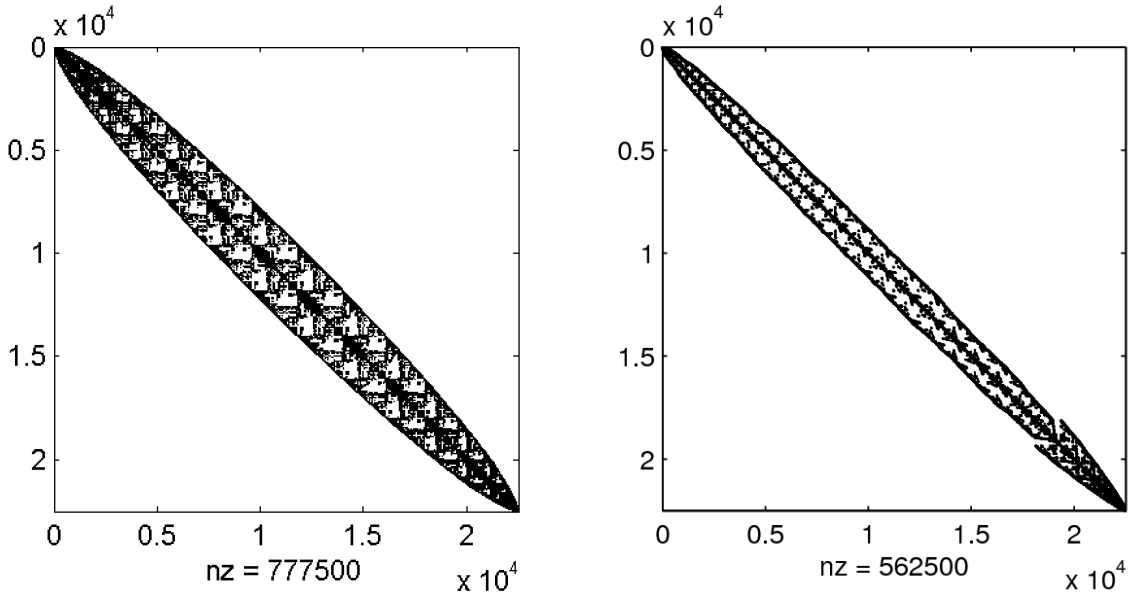


Figure 10: Schematic picture of the two types of stencils. Left panel: Explicit stencil. Right panel: Implicit stencil.

reverse Cuthill–McKee reordering of the entries, is shown in Figure 11. The horizontal resolution is given by N_H, and the radial one by N_R. The total number of nodes is denoted $N = N_H \cdot N_R$.

The linear system arising from the discretization is solved using BiCGSTAB with zero-fill modified incomplete LU-factorization as a preconditioner. A residual tolerance of 10^{-8} is used in the iterative solver. The subplots in Figure 12 show the relative ℓ_2 error and the number of iterations required, respectively, as a function of the total number of nodes. While both methods achieve 4th order convergence, the implicit method is more efficient in terms of memory and number of iterations required to converge.

Table 5 gives the exact specifications of the 6 node sets used for the results in Figure 12.



(a) Explicit stencil

(b) Implicit stencil

Figure 11: The sparsity pattern for $N_H = 2500$, $N_R = 11$.

5.2 Mantle convection

Thermal convection of an incompressible, Boussinesq fluid at infinite Prandtl number in a spherical shell that is heated from below is discussed here. This can be considered as a simplified version of mantle convection with constant viscosity. The dynamics of the fluid are governed by the Rayleigh number, Ra, which can be interpreted as a ratio of the destabilizing force due to the buoyancy of the heated fluid to the stabilizing force due to the viscosity of the fluid. Since the fluid is incompressible and isoviscous the velocity can be be expressed solely in terms of a poloidal potential, $\mathbf{u} = \nabla \times \nabla \times ((\Phi r)\hat{\mathbf{r}})$ (see, for example, [3; 8]). Exploiting this relationship, the governing equations for a shell of inner radius R_i and outer radius R_o can be written as

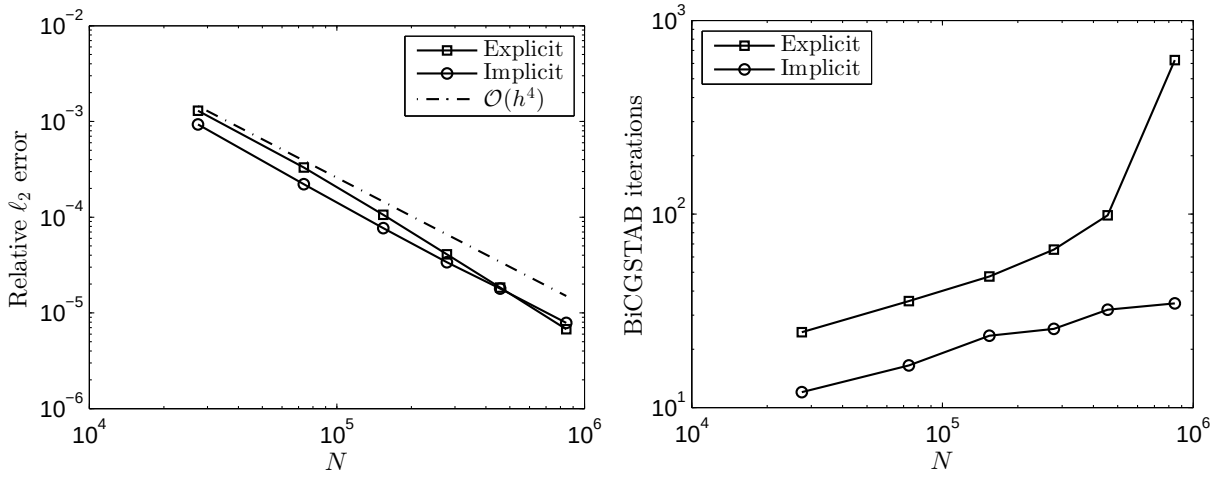
$$\Delta_{\mathcal{S}}\Omega + \frac{\partial}{\partial r}\left(r^2\frac{\partial \Omega}{\partial r}\right) = \mathrm{Ra}\, r\, T, \tag{24}$$

$$\Delta_{\mathcal{S}}\Phi + \frac{\partial}{\partial r}\left(r^2\frac{\partial \Phi}{\partial r}\right) = r^2\Omega, \tag{25}$$

$$\frac{\partial T}{\partial t} = -\left(u_r\frac{\partial T}{\partial r} + u_\theta\frac{1}{r}\frac{\partial T}{\partial \theta} + u_\lambda\frac{1}{r\cos\theta}\frac{\partial T}{\partial \lambda}\right) + \frac{1}{r^2}\Delta_{\mathcal{S}}T + \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial T}{\partial r}\right), \tag{26}$$

where $R_\mathrm{i} \le r \le R_\mathrm{o}$, $-\pi/2 \le \theta \le \pi/2$, $-\pi < \lambda \le \pi$ are the standard spherical coordinates, $\mathbf{u} = (u_r, u_\theta, u_\lambda)$ is the velocity field in spherical coordinates, T is temperature, and $\Delta_{\mathcal{S}}$ is the surface Laplacian operator. The boundary conditions on the velocity of the fluid at the inner and outer surfaces of the spherical shell are assumed to be impermeable and shear-stress free, which translates into

$$\Phi|_{r=R_\mathrm{i},R_\mathrm{o}} = 0 \qquad \text{and} \qquad \left.\frac{\partial^2\Phi}{\partial r^2}\right|_{r=R_\mathrm{i},R_\mathrm{o}} = 0. \tag{27}$$



(a) Error (b) Iterations

Figure 12: The error and the number of iterations as a function of N.

Table 5: The specifications for the results shown in Figure 12.

Stencil type	N_R	N_H	ε_R	ε_H	ℓ_2 error	Iterations
Explicit	11	2500	0.5	5	0.00129	24.5
	15	4900			0.000330	35.5
	19	8100			0.000106	47.5
	23	12100			$4.09 \cdot 10^{-5}$	65.5
	27	16900			$1.84 \cdot 10^{-5}$	98.5
	33	25600			$6.73 \cdot 10^{-6}$	624.5
Implicit	11	2500	0.5	4	0.000930	12
	15	4900			0.000221	16.5
	19	8100			$7.68 \cdot 10^{-5}$	23.5
	23	12100			$3.39 \cdot 10^{-5}$	25.5
	27	16900			$1.79 \cdot 10^{-5}$	32
	33	25600			$7.87 \cdot 10^{-6}$	34.5

The boundary conditions on the temperature are

$$T(R_{\mathrm{i}}, \theta, \lambda) = 1 \quad \text{and} \quad T(R_{\mathrm{o}}, \theta, \lambda) = 0.$$

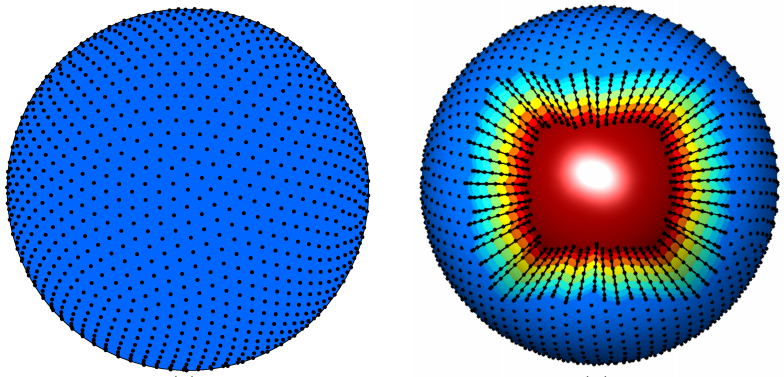
Equations (24)-(26) have been non-dimensionalized with the length scale chosen as the thickness of the shell, $\Delta R = R_{\mathrm{o}} - R_{\mathrm{i}}$, the time-scale chosen as the thermal diffusion time, $t = (\Delta R)^2/\kappa$ (κ=thermal diffusivity), and the temperature scale chosen as the difference between the temperature at the inner and outer boundaries, ΔT. In the experiments below, the inner and outer radii of the shell are set to $R_{\mathrm{o}} = 20/9$ and $R_{\mathrm{i}} = 11/9$, which give approximately the same ratio of the inner and outer radii of the Earth's mantle (i.e. 0.55).

5.2.1 Overview of the numerical approach

To numerically solve (24)–(26) an operator-splitting method similar to [61] is used in space where the lateral directions (θ, λ) are discretized separately from the radial direction; see Figure 13 for an

illustration. M Chebyshev nodes are used in the radial direction and N "scattered" nodes are used on each of the resulting M spherical surfaces, giving a total of MN nodes in the 3-D spherical shell.

The maximal determinant (MD) nodes [59] are again used for the spherical surfaces. All radial derivatives are discretized using collocation with Chebyshev polynomials (see, for example, [14; 52] for details), while all differential operators in the latitudinal direction θ and longitudinal direction λ are approximated on each spherical surface using n-node RBF-FD stencils.



(a) (b)

Figure 13: (a) Node layout for the RBF-FD approximations on the surface of a sphere; (b) 3-D view of the discretization of the spherical shell used in the hybrid RBF-FD/Chebyshev calculation. Blue is the outer boundary and red is the inner boundary, black circles display the computational nodes, which are distributed in the radial direction along the extrema of the Chebyshev polynomials. The spherical shell has been opened up in (b) to show the detail of the radial discretization.

The complete numerical algorithm is nearly identical to the one in [61], with the only difference being that RBF-FD is used instead of global RBFs. The reader is therefore referred to Appendix B of [61] for a detailed description. A general overview is given by the following steps:

1. Discretize $\dfrac{\partial}{\partial\theta}$, $\dfrac{1}{\cos\theta}\dfrac{\partial}{\partial\lambda}$, and $\Delta_{\mathcal{S}}$ at each of the N MD nodes using n point RBF-FD formula. As discussed in [61], the coordinate singularities associated with the latter of these two operators is harmlessly removed when applied to an RBF expansion.

2. Discretize $\dfrac{\partial}{\partial r}$, $\dfrac{\partial}{\partial r}\left(r^2\dfrac{\partial}{\partial r}\right)$, $\left.\dfrac{\partial^2}{\partial r^2}\right|_{r=R_{\text{i}}}$, and $\left.\dfrac{\partial^2}{\partial r^2}\right|_{r=R_{\text{o}}}$ using collocation with M Chebyshev polynomials.

3. With the given temperature initial condition, solve the Poisson equations (24) and (25) for Ω and Φ. This procedure is complicated by the fact that all 4 boundary conditions on these equations are only specified on Φ. To handle this, the influence matrix [41] method is employed to find the unknown boundary values on Ω such that all 4 boundary conditions on Φ are satisfied (see Section 5 of [61]).

4. Compute the velocity field: $\mathbf{u} = \nabla\times\nabla\times[(\Phi r)\hat{\mathbf{r}}] = \left(\dfrac{1}{r}\Delta_{\mathcal{S}}\Phi, \dfrac{1}{r}\dfrac{\partial^2}{\partial r\partial\theta}(\Phi r), \dfrac{1}{r\cos\theta}\dfrac{\partial^2}{\partial r\partial\lambda}(\Phi r)\right)$.

5. Update the energy equation (26) to the next time step $t+\Delta t$ using the semi-implicit third order Adams-Bashforth (AB3) and second order trapezoidal rule (or Crank-Nicolson) method.

In this case, all terms are treated explicitly with AB3, except the radial component of the diffusion term which is treated implicitly using trapezoidal rule. Return to Step 3 with an updated temperature profile.

5.2.2 Validation on community benchmark problems

The two most common benchmarks for computational models of mantle convection in a spherical shell are the isoviscous steady-state tetrahedral and cubic test cases. These names correspond to the steady-state upwelling plumes that result from these tests at the faces of regular tetrahedron and cube, respectively. The initial condition for the temperature is specified as

$$T(r,\theta,\lambda) = \frac{R_{\mathrm{i}}(r-R_{\mathrm{o}})}{r(R_{\mathrm{i}}-R_{\mathrm{o}})} + 0.01 Y_3^2(\theta,\lambda)\sin\left(\pi\frac{r-R_{\mathrm{i}}}{R_{\mathrm{o}}-R_{\mathrm{i}}}\right) \tag{28}$$

for the tetrahedral test case and

$$T(r,\theta,\lambda) = \frac{R_{\mathrm{i}}(r-R_{\mathrm{o}})}{r(R_{\mathrm{i}}-R_{\mathrm{o}})} + 0.01\left[Y_4^0(\theta,\lambda) + \frac{5}{7}Y_4^4(\theta,\lambda)\right]\sin\left(\pi\frac{r-R_{\mathrm{i}}}{R_{\mathrm{o}}-R_{\mathrm{i}}}\right) \tag{29}$$

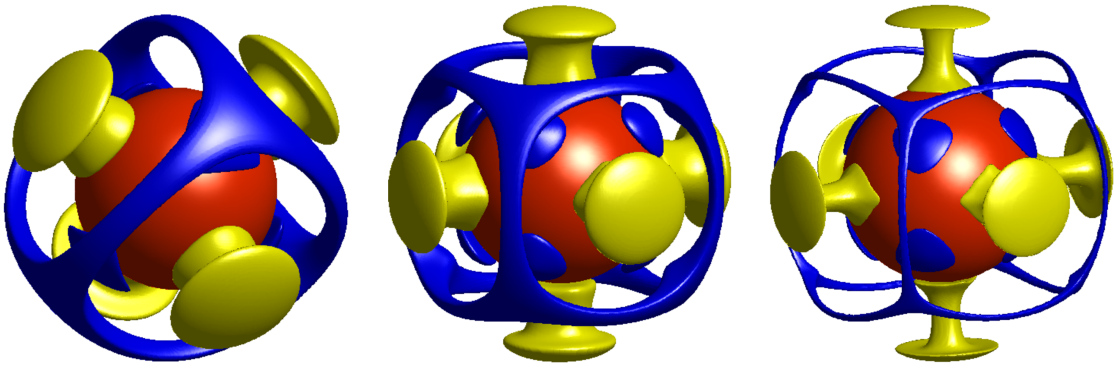
for the cubic test case, where Y_ℓ^m denotes the normalized spherical harmonic of degree ℓ and order m. The first term in each of the initial conditions represents a purely conductive temperature profile, while the second terms are perturbations to this profile and determine the final steady-state solution.

Both the tetrahedral and cubic tests were first run at $\mathrm{Ra} = 7000$, with $N = 2601$ nodes on each spherical shell and $M = 23$ Chebyshev nodes in the radial direction giving a total of $59,823$ nodes. Each differential operator on the spherical surfaces was approximated using $n = 50$ node RBF-FD formulas. A time-step of 10^{-4} was used to reach steady-state at the non-dimensionalized time of $t = 1$, corresponding to roughly 58 times the age of the Earth. The final RBF-FD steady-state solutions for the Tetrahedral and Cubic test cases are displayed in Figures 14(a) and (b), respectively. As no analytical solutions exist, validation is done via comparison to other published results in the literature with respect to scalar global quantities, such as Nusselt number at the inner and outer boundaries ($\mathrm{Nu_i}$ and $\mathrm{Nu_o}$), and the averaged root mean square velocity and temperature over the volume. Such a comparison for the RBF-FD method with respect to established methods (including the global RBF-PS method [61]) is given in Table 6. For the RBF-FD and RBF-PS methods, the standard deviation of all the quantities from the last 1000 time-steps was less than $5 \cdot 10^{-5}$, which is a standard measure for indicating the model has reached numerical steady-state.

While the $\mathrm{Ra} = 7000$ tests above are the *de facto* benchmarks, there are also some results in the literature for the cubic test at $\mathrm{Ra} = 10^5$. This Rayleigh number results in a more convective regime, with thinner plumes, so that higher resolutions are needed to properly capture the solution. For this test, $N = 6561$ nodes were used on each spherical shell and $M = 43$ Chebyshev nodes in the radial direction. The spherical surface differential operators were again approximated using $n = 50$. Increasing M necessitates a decrease in the time-step to maintain time-stability since the Chebyshev nodes cluster quadratically at the boundaries. A time-step of $\Delta = 6 \cdot 10^{-6}$ was used to reach a final time of $t = 0.35$, at which numerical steady-state was reached (using the same criteria as the $\mathrm{Ra} = 7000$ cases). The final RBF-FD steady-state solution is displayed in Figures 14(c), and a comparison with two other methods is given in Table 7.

The following observations can be made regarding the results for these benchmarks:

1. For the $\mathrm{Ra} = 7000$ test case, the results from the RBF-PS method [61] and Harder's extrapolated results from a spherical harmonic-finite difference method [31] are expected to be



(a)Tetrahedral, Ra $= 7 \cdot 10^3$ (b) Cubic, Ra $= 7 \cdot 10^3$ (c) Cubic, Ra $= 1 \cdot 10^5$

Figure 14: Steady-state isosurfaces of the residual temperature, δT, for the three isoviscous benchmark test cases using the RBF-FD model. Here the residual temperature $\delta T = T(r, \theta, \lambda) - \langle T(r) \rangle$ is visualized, where $\langle\ \rangle$ denotes averaging over a spherical surface. Yellow corresponds to $\delta T = 0.15$ and denotes upwelling relative to the average temperature at each radial level, while blue corresponds to $\delta T = -0.15$ and denotes downwelling. The red solid sphere shows the inner boundary of the 3-D shell corresponding to the core.

correct to 4 digits. The hybrid RBF-FD method was able to match these values to four digits for both tests. Furthermore, only in three values does the RBF-FD method not match the global RBF-PS method to all digits.

2. The number of nodes (degrees of freedom) needed to accomplish the Ra $= 7000$ and Ra $= 10^5$ results for the RBF-FD method are significantly lower (over an order of magnitude in all but one case) than any of the other methods that use a hybrid spectral-FD discretization or a full finite difference, finite volume, or finite element method.

3. The Nusselt number is non-dimensional, measuring the ratio of convective to conductive heat transfer across a boundary. Thus if there are no sources or sinks in the domain, energy should be conserved and $\mathrm{Nu_i} = \mathrm{Nu_o}$. This is exhibited quite clearly in the Ra $= 7000$ cases for the RBF-FD method (like the RBF-PS), and to a slightly lesser extent in the Ra $= 10^5$ (with a difference of only 0.05%). This suggests the method will inherently dissipate physical quantities less.

4. While the RBF-FD method requires a higher resolution (larger N) than the global RBF-PS method to achieve similar results, it has a significant advantage in terms of computational cost and memory storage, as spatial discretizations require a small fraction of the number of terms required by the global method.

5.2.3 Fully convective flow: Ra $= 10^6$

The final results for this section are from a simulation of the isoviscous mantle convection model with Ra $= 10^6$, which is well past the critical threshold to put the flow in a fully convective state. The initial condition for the simulation was obtained from a simulation at a lower Ra that was initialized using a purely conductive temperature profile plus a small perturbation in the lateral direction that randomly combined all spherical harmonics up to degree 10 (see [61] for more details on this "spin-up" procedure). For the discretization, $M = 81$ Chebyshev nodes were used in the radial direction (to resolve the thin boundary layers) and $N = 10000$ nodes were used on

Table 6: Comparison between computational methods for the isoviscous Tetrahedral and Cubic mantle convection test cases with Ra = 7000. $\mathrm{Nu_o}$ and $\mathrm{Nu_i}$ denote the respective Nusselt number at the outer and inner spherical surfaces, $\langle V_{\mathrm{rms}} \rangle$ the volume-averaged RMS-velocity over the 3-D shell, and $\langle T \rangle$ the mean temperature of the 3-D shell. Extrap. indicates that the results were obtained using Romberg extrapolation. Solid lines indicate numbers were not reported. Abbreviations: FE = finite element, FD = finite difference, FV = finite volume, SP-FD = hybrid spectral and finite difference, and SP = purely spectral.

Model	Type	Nodes	$r \times (\theta \times \lambda)$	$\mathrm{Nu_o}$	$\mathrm{Nu_i}$	$\langle V_{\mathrm{rms}} \rangle$	$\langle T \rangle$
Tetrahedral test case, Ra = 7000							
Zhong [65]	FE	393216	$32 \times (12 \times 32 \times 32)$	3.5126	3.4919	32.66	0.2171
Yoshida [63]	FD	2122416	$102 \times (102 \times 204)$	3.4430	—	32.0481	—
Kameyama [34]	FD	12582912	$128 \times (2 \times 128 \times 384)$	3.4945	—	32.6308	0.21597
Ratcliff [44]	FV	200000	$40 \times (50 \times 100)$	3.4423	—	32.19	—
Stemmer [50]	FV	663552	$48 \times (6 \times 48 \times 48)$	3.4864	3.4864	32.5894	0.21564
Stemmer [50]	FV	Extrap.	Extrap.	3.4949	—	32.6234	0.21560
Harder [31; 50]	SP-FD	552960	$120 \times (48 \times 96)$	3.4955	—	32.6375	0.21561
Harder [31; 50]	SP-FD	Extrap.	Extrap.	3.4962	—	32.6424	0.21556
RBF-PS [61]	SP	36800	$23 \times (1600)$	3.4962	3.4962	32.6424	0.21556
RBF-FD, $n = 50$	SP-FD	59823	$23 \times (2601)$	3.4962	3.4962	32.6425	0.21556
Cubic test case, Ra = 7000							
Zhong [65]	FE	393216	$32 \times (12 \times 32 \times 32)$	3.6254	3.6016	31.09	0.2176
Yoshida [63]	FD	2122416	$102 \times (102 \times 204)$	3.5554	—	30.5197	—
Kameyama [34]	FD	12582912	$128 \times (2 \times 128 \times 384)$	3.6083	—	31.0741	0.21639
Ratcliff [44]	FV	200000	$40 \times (50 \times 100)$	3.5806	—	30.87	—
Stemmer [50]	FV	663552	$48 \times (6 \times 48 \times 48)$	3.5983	3.5984	31.0226	0.21594
Stemmer [50]	FV	Extrap.	Extrap.	3.6090	—	31.0709	0.21583
Harder [31; 50]	SP-FD	552960	$120 \times (48 \times 96)$	3.6086	—	31.0765	0.21582
Harder [31; 50]	SP-FD	Extrap.	Extrap.	3.6096	—	31.0821	0.21578
RBF-PS [61]	SP	36800	$23 \times (1600)$	3.6096	3.6096	31.0820	0.21577
RBF-FD, $n = 50$	SP-FD	59823	$23 \times (2601)$	3.6095	3.6096	31.0819	0.21577

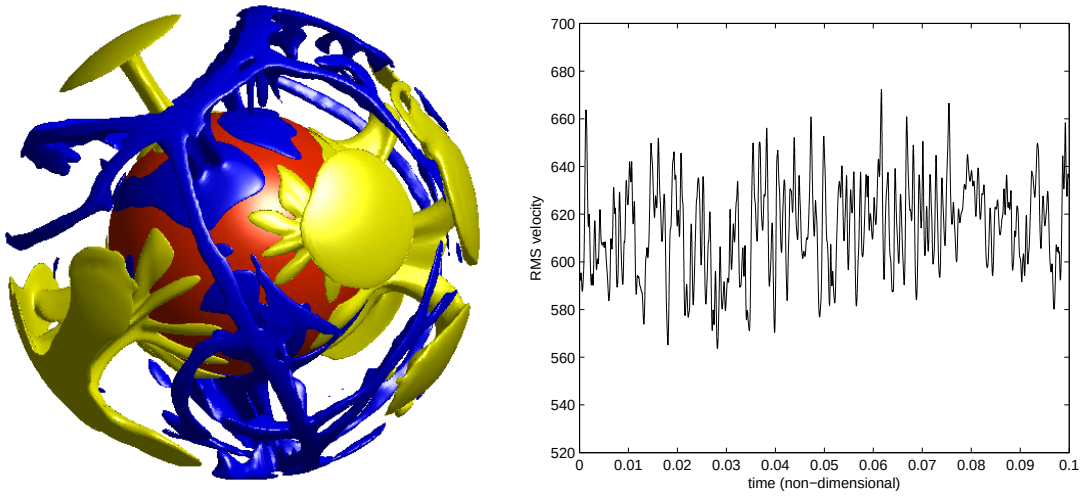
each spherical shell, with $n = 50$ node RBF-FD formulas for approximating the spherical surface differential operators. A hyperviscosity filter of order $k = 6$ (cf. Section 2.4) was also used in the numerical solution of the energy equation to stabilize the time-integration. The simulation was run from $t = 0$ to $t = 0.1$ (approximately 5 times the age of the Earth). Figure 15(a) shows the isosurfaces of the residual temperature at $t = 0.09$ and clearly shows that plumes with very fine structure are present. Figure 15(b) shows a time-trace of the volume averaged RMS-velocity, $\langle V_{\mathrm{rms}} \rangle$, over the entire simulation, which illustrates that the mantle is in a fully convective regime. At present, this is the largest simulation using RBF-FD in spherical geometry, with nearly 1 million degrees of freedom.

6 Future directions and concluding remarks

This article provides a survey of a group of topics relevant to using RBF-FD, a mesh-less method, for a variety of problems that arise in the geosciences, with particular emphasis on spherical geometries, both on surfaces and within a volume. RBF-FD is a method that has only begun to grow in the last few years. From a practical standpoint, RBF-FD sprouted out of global RBFs, which showed exceptional numerical qualities in terms of accuracy and time stability for solving PDEs

Table 7: Comparison between computational methods for the isoviscous Cubic mantle convection test case with $\mathrm{Ra} = 10^5$.

Model	Type	Nodes	$r \times (\theta \times \lambda)$	$\mathrm{Nu_o}$	$\mathrm{Nu_i}$	$\langle V_{\mathrm{rms}} \rangle$	$\langle T \rangle$
Zhong [65]	FE	1,327,104	$48 \times (12 \times 48 \times 48)$	7.8495	7.7701	154.8	0.1728
RBF-PS [61]	SP	176,128	$43 \times (4096)$	7.8120	7.8005	154.49	0.17123
RBF-FD, $n = 50$	SP-FD	282,123	$43 \times (6561)$	7.8072	7.8030	154.43	0.17069



(a) (b)

Figure 15: Results from a fully convective simulation with $\mathrm{Ra} = 10^6$. (a) Shows the isosurfaces of the residual temperature, δT, at $t = 0.09$, where yellow corresponds to $\delta T = 0.1$, blue corresponds to $\delta T = -0.1$, and the red solid sphere shows the inner boundary of the 3-D shell corresponding to the core. (b) Shows the average RMS velocity, $\langle V_{\mathrm{rms}} \rangle$, over the complete simulation.

but reduction in computational cost needed to be addressed if RBFs were to be effective when scaled to very large problem sizes. With the RBF-FD methodology, matrices go from full to becoming 99% empty. Of course, the sacrifice is the exchange of spectral accuracy for high-order algebraic convergence, assuming smooth data. However, since natural processes are almost never infinitely differentiable, little is lost and much gained in terms of memory and runtime.

Although the future of RBF-FD is bright on the horizon, many topics still need to be addressed. These include but are definitely not limited to improved reliability, effectiveness, and scalability of RBF-FD implementations on novel computer architectures [6], dynamic adaptive node refinement [11], application of preconditioners and iterative solvers for elliptic PDEs, stability analysis in the presence of boundary conditions for hyperbolic problems, and the treatment of discontinuities in the domain.

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